

GEMP ACCESS COURSE: BIOLOGICAL CHEMISTRY

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Lecture 3: Atom and Atomic Structure

Kwame Nkrumah University of Science and Technology, Kumasi | Leaders In Change

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Dalton's Atomic Theory

- Each element is made up of unique type of atoms which are indivisible.
- Atoms can neither be created nor destroyed.
- Atoms of the same element are identical: they have identical mass and show the same physical and chemical properties.
- When elements combine to form compounds, their atoms join together in fixed proportions.



Various models of atomic structure

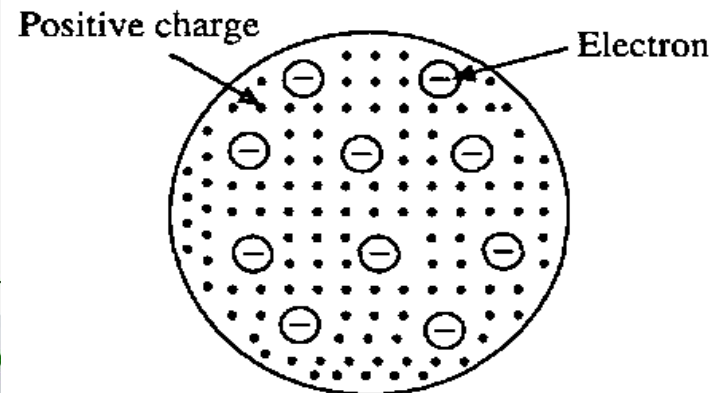
The following will be considered:

- The plum pudding (watermelon) model of J.J. Thomson.
- The nuclear atom model of Rutherford
- The planetary model of Bohr.
- The wave mechanics model of Schrodinger.



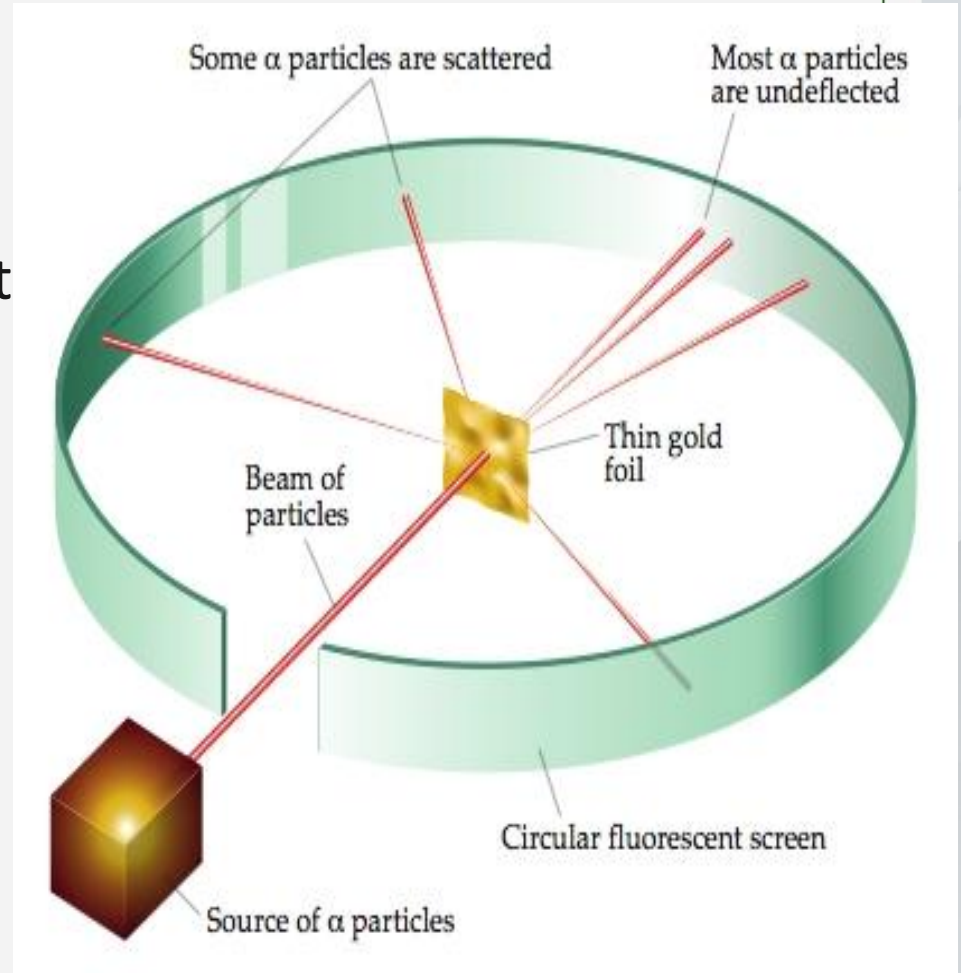
The plum-pudding model

- The positive charges (protons) of an atom are uniformly distributed in a sphere
- The electrons are also embedded in the sphere in such a way that the attraction of the electrons for the positive charge just overcomes the repulsion among the electrons.
- Thomson's model of an atom had no nucleus.



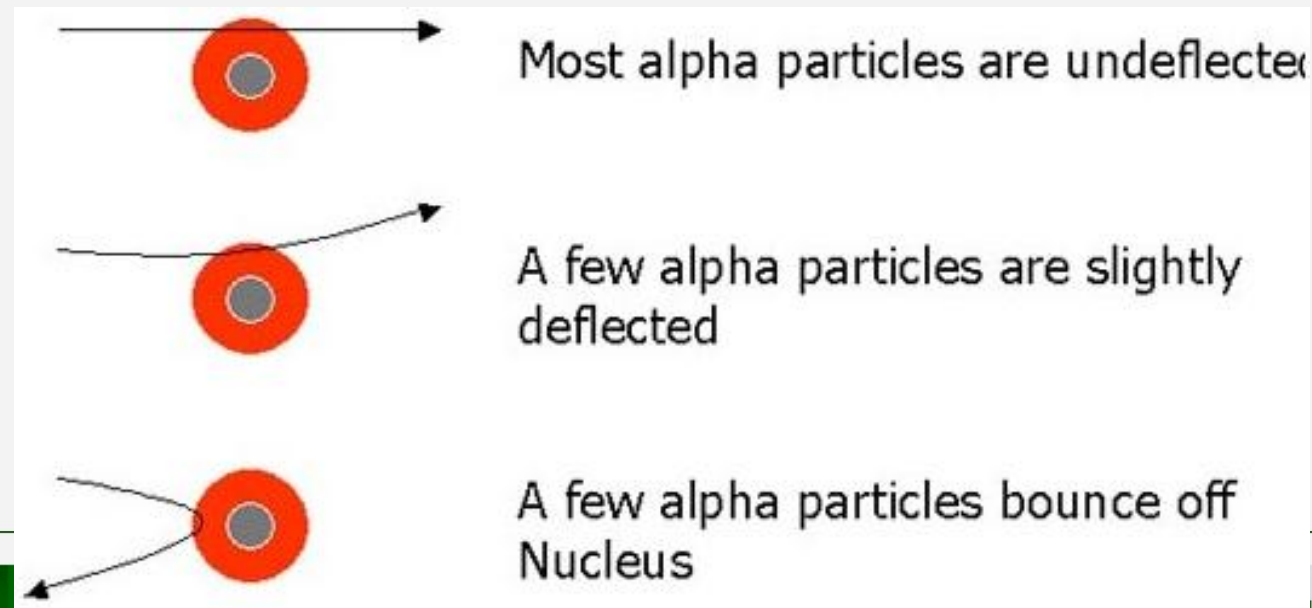
Rutherford's α -scattering experiment

- Rutherford, a student of Thomson set out to prove the plum-pudding model of the atom his mentor had proposed.
- Rutherford assigned the experiment to his assistant Hans Geiger and undergraduate student, Ernest Marsden.
 - The experiment is also called the gold foil experiment.
 - A particle gun (containing radon) which could emit α -particle was put before a thin sheet of gold.
 - Then a detector was placed beyond the gold foil.
 - The detector was a zinc sulphide screen which could fluoresce as the alpha-particles hit it.



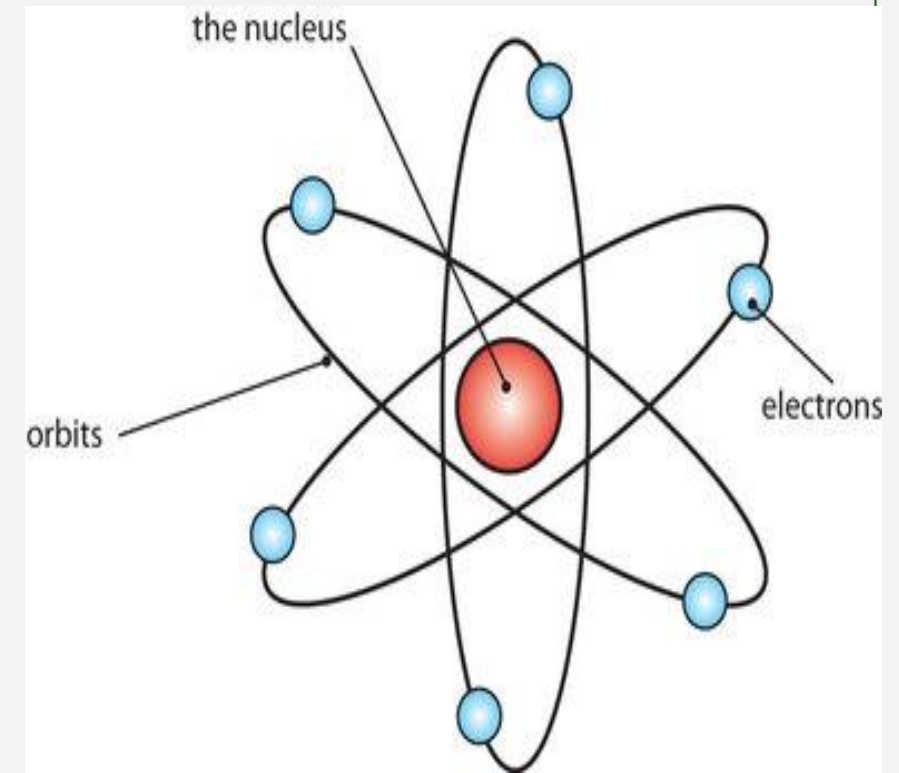
Observations from Rutherford's experiment

- Most of the α -particles went through the gold foil undeflected.
- A small percentage of the α -particles were deflected through some small angles.
- Very few of the particles bounced back or were deflected through large angles.



Rutherford's nuclear atom

- From the α -scattering experiment, Rutherford could deduce the structure of an atom.
 - That an atom has a central nucleus containing the positively charged protons.
 - The mass of the atom is concentrated in the nucleus, but the volume of the nucleus was relatively small as compared to the volume occupied by the electrons.
 - It was also held that electrons move speedily around the nucleus in circular orbits.

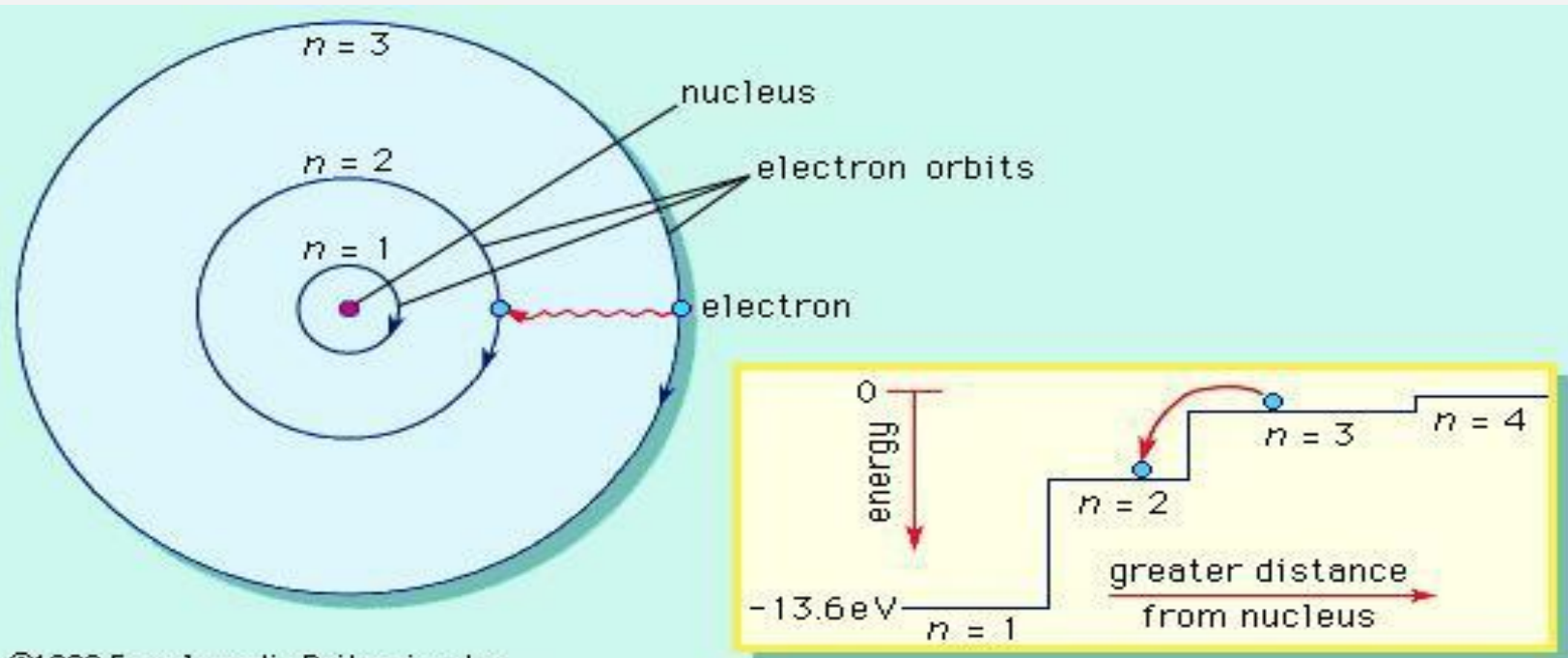


Bohr's model of an atom

- From Rutherford's model of the atom it was thought that electrons orbit around the nucleus at high speed.
 - However, scientists wondered why the negative electrons did not lose energy and spiral into the nucleus.
- Why nothing like that happened was taken care of by the quantum theory, of which Niels Bohr, a Danish physicist was a chief exponent.
 - Using the hydrogen atom for his study, Bohr suggested that the electron in hydrogen could not just have any energy, but that the electron was restricted to some permissible energy levels.
 - Thus he talked of the fact that the electrons were in discrete energy levels.



- What facilitated Bohr's study of hydrogen atom was his examination of spectra.
- When hydrogen atoms are given energy by passing an electric current through it at low pressure, or when it is heated in a flame, a line spectrum was observed.



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- A spectrophotometer is a device for producing and observing spectra.
 - It is made up of an energy source (heat or electric), a prism or diffracting grating to separate the light and an eyepiece.
 - When different elements are heated in a flame, they glow, and upon viewing the glowing light in a spectroscope, depending on the type of element, some series of brightly coloured lines are seen separated by dark lines.
- The spectra of different elements are different.
 - The spectrum of sodium has two closely-spaced yellow lines at 590 and 589 nm.
 - The spectrum of hydrogen is made up of red, green and purple lines at 656, 486 and 434 nm respectively.



- Contrast to the line spectrum of elements with a continuous spectrum, which is obtained when sunlight is passed through a prism.
 - In the latter case, light is separated into a rainbow of colours.
 - This is a continuous spectrum in which one colour of light fuses into another. Such a spectrum contains light of all wavelengths.
- Niels Bohr explained that electrons were responsible for the line spectrum of hydrogen of the way electrons are arranged in energy levels around the nucleus.
 - The electrons were said to revolve around the nucleus in definite orbits or energy levels like planets revolve around the sun. Each of the orbits has a certain energy associated with it.
 - Only certain energy values were permissible for these orbits.
 - Therefore, the energy of the electron was quantized.



- In Bohr's model, each orbit was assigned a number called the **principal quantum number**.
- This could be a whole number from 1 to infinity.
- When the electron is in the lowest energy level or orbit, it is said to be in the ground state.
- Electrons can move from a lower energy state to a higher one if energy is supplied to it.
 - By this the electrons become excited.
 - The electron will not remain in the excited state indefinitely.
 - It can lose some or all the energy absorbed, and this lost energy is emitted as photon of light which could be visible or invisible.



- The decrease in energy can be designated as ΔE , where

$$\Delta E = h\nu \text{ or } \Delta E = nh\nu,$$

- h = Planck's constant
 - ν = frequency of the radiation
 - n = integer
-
- If the decrease in energy of the atom is large then a photon of light of high frequency will be emitted which may be in the blue or UV region.
 - On the other hand, if the decrease in energy is low, then a photon of light of low frequency would be emitted, maybe in the red or infra-red region of the electro-magnetic spectrum.



Quantum or Wave Mechanics Model of An Atom

- The Bohr's atomic model could not account for the line spectra of elements bigger than hydrogen.
 - The appearance of the spectra of such bigger atoms had more complicated pattern of lines, which could not be accounted for.
- The Bohr model considered an electron to be particle but Louis de Broglie, a French graduate student suggested in 1924 that apart from having a particle nature, electrons also had wave properties.
 - He formulated an equation which took account of this wave-particle dual nature of electron.



- As a particle, it would have a momentum associated with it, while as a wave, it would have a wavelength.
- The relationship between the mass of the electrons, its wavelength and velocity (v) was shown as:
$$\lambda = \frac{nh}{mv}$$
- As soon as scientists began to consider an electron as behaving as waves, a problem arose:
 - for a subatomic particle, would it be possible to know its position and momentum or speed?



- In answer to this, a German physicist, Werner Heisenberg postulated the 'uncertainty principle'.
 - It is difficult to define the position and momentum of an electron at the same time.
- The Bohr's idea of an atom violated this principle because both the position of the electron and its energy was known.
 - The line spectrum indicated that electrons were in definite orbits with definite energies.
 - To be in consonance with the uncertainty principle, the fact that electrons have definite energies is upheld.
 - It is rather the position of the electron that cannot be known with any degree of certainty.



- Erwin Schrodinger was to combine de Broglie's equation with other equations for wave motion to derive a new wave equation or Schrodinger equation to describe the behaviour of an electron in a hydrogen atom.
 - For atoms with more than one electron, the calculations were even more complicated.
- Solutions of the Schrodinger equation are functions called wave functions psi (Ψ).
 - The square of the wave function (Ψ^2) is a mathematical expression of how the probability of finding an electron in a small volume varies from place to place.
 -
 - In atoms the wave functions are called **orbitals**.
 - An orbital can be defined as a volume of space around the nucleus within which there is 95% probability of finding an electron.



- The Bohr model also considered only one quantum number, but in the wave mechanics model, two other quantum numbers,
 - the orbital or azimuthal or shape quantum number (ℓ) and
 - magnetic quantum numbers (m_ℓ),
- in addition to the principal quantum number (n) are considered.



The Principal quantum number (n)

- It has integral values from 1 to infinity.
 - It can also be represented by the letters, K, L, M, N...
- The value of n gives an indication of how close an electron is to the nucleus.
 - The higher the value of n , the farther away the electron is from the nucleus.
- The value of n also determines the energy of the electron.
 - The higher the value, the higher the energy of the electron.
- For a principal quantum number n , the maximum number of electrons it can accommodate is given by $2n^2$.



- The table below shows the maximum number of electrons in the first four shells,

n	Max. no. of electrons
1	2
2	8
3	18
4	32

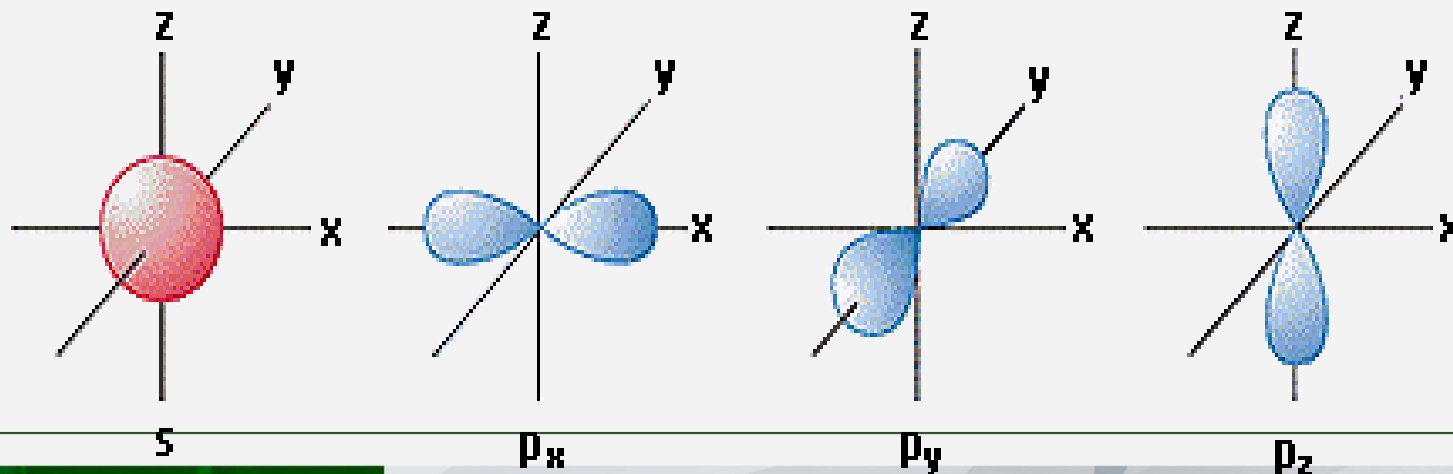


The orbital or azimuthal or angular momentum quantum number (l)

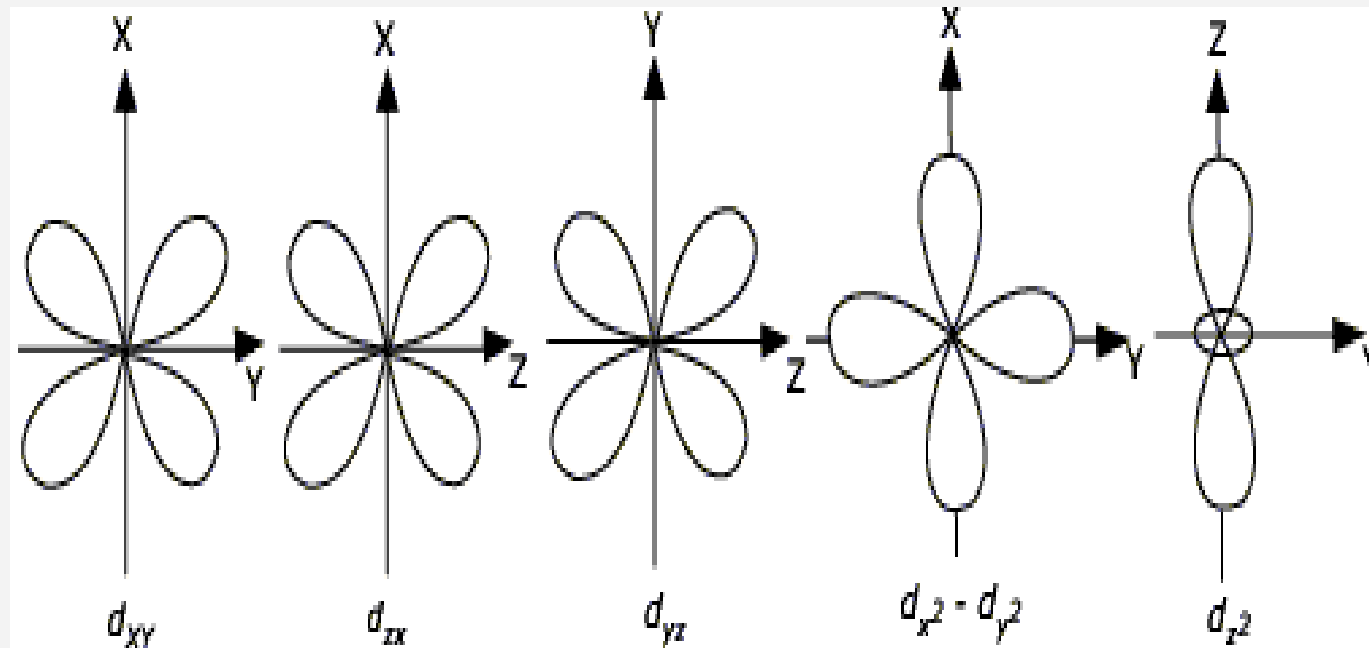
- It can have values from zero to $n-1$.
- This quantum number defines the shape of an orbital.
 - Thus there is only one orbital or subshell in the first shell of $n=1$.
- This orbital with $l=0$ is also called the s-orbital.
 - The shape of an s-orbital is symmetrically spherical.
 - All s-orbitals have the same shape, and each orbital can take two electrons.



- When $n = 2$; $l = 0, 1$ or s and p-orbitals.
- Thus the shell of principal quantum number 2 has two orbitals or subshells, 2s and 2p.
- The shape of a p-orbital is different from that of an s-orbital.
- Whereas the s-orbital is spherical, the p-orbital is dumb-bell in shape.



- The shell with principal quantum number 3 will have three orbitals; s, p and d.
- There will be 3s, 3p and 3d orbitals.
- The shapes of the d-orbitals are more complex.



The magnetic quantum number (m_l)

- It can have whole number values of $-l$ to $+l$, including zero.
- It describes the orientation of the orbitals in space.
- When $n = 1$, $l = 0$.
 - Thus, the s-orbital has no directions of orientation.
- When $n = 2$, $l = 0, 1$ $m_l = -1, 0, +1$.
 - Thus there are three p-orbitals of the same shape but differ with respect to their orientation in space.
 - Each of the three can be arbitrarily assigned to one of the Cartesian x , y and z .
 - So the three are mutually perpendicular.



- When $n=3$, $l = 0,1,2$; $m_l = -2, -1, 0, 1, +2$
 - The d-subshell is made up of five orbitals. So in all, ten electrons can be accommodated in the d-subshell while the p-subshell takes six electrons.
- The p-subshell has three orbitals of the same energy or degenerate orbitals while the d-subshell has five degenerate orbitals.



Spin quantum number

- This is the fourth type of quantum number.
- It arises from the ability of electrons to spin about their axes either in the clockwise or anti-clockwise direction.
- The spin quantum numbers have values of $+\frac{1}{2}$, $-\frac{1}{2}$.



Electronic configuration

- This is the arrangement of electrons in atoms.
 - It is the specific pattern of arrangement that will make every element unique.
- In a way the electronic configuration of an element will serve as the 'face' for the identification of the elements.
- There can be two approaches for arranging electrons in elements.
 - The first is simple in that the electrons are arranged in shells, taking into account only the principal quantum number, n .
 - The second approach which is rather more involving requires the consideration of all the four quantum numbers.



- Whichever of the two approaches is used, there is one rule that has to be adhered to - the 'aufbau rule'.
 - According to this rule, electronic energy levels have to be filled starting from the low energy level.
 - Thus, we have to start from $n = 1$ before proceeding to $n = 2$, and so on.
- For the simplest approach, we could either draw the energy levels as concentric circles around the nucleus or we could just represent the energy levels by the letters, K, L, M, N



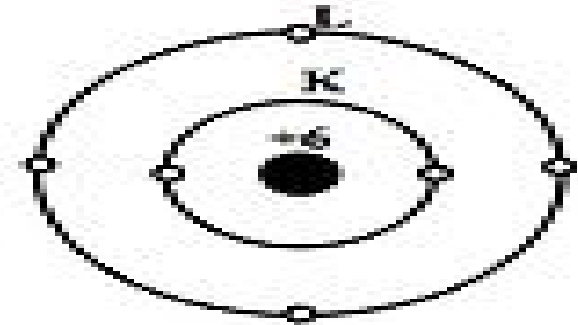
- Bear in mind that each shell has the maximum number of electrons it can take.



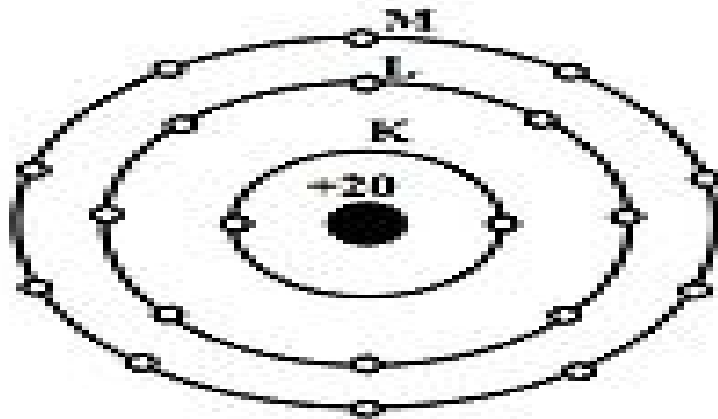
Hydrogen



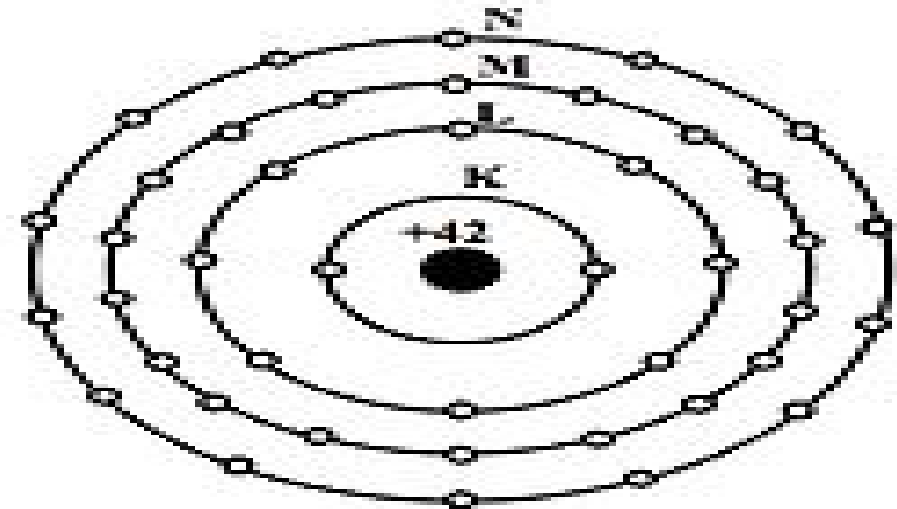
Helium



Carbon



Calcium



Molybdenum



Atomic no.	Symbol	K	L	M	N
1	H	1			
2	He	2			
3	Li	2	1		
4	Be	2	2		
11	Na	2	8	1	
12	Mg	2	8	2	
13	Al	2	8	3	
14	Si	2	8	4	
19	K	2	8	8	1
20	Ca	2	8	8	2
21	Sc	2	8	9	2
28	Ni	2	8	16	2
29	Cu	2	8	18	1
30	Zn	2	8	18	2

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- In the other way of arrangement of electrons, atomic orbitals in the various shells are used.
 - According to the aufbau rule, the order of filling of the orbitals is as follows:
 - 1s 2s 2p 3s 3p 4s 3d 4p
- Two other rules are considered;

Hunds rule

- In filling degenerate orbitals, they are filled singly with electrons to maximum before pairing up the electrons in the orbitals.

Pauli' s exclusion principle

- Pauli's exclusion principle states that no two electrons in an orbital would have all the four numbers being the same.



		Orbitals			
		s	p	d	f
Principle Quantum Number (Energy Level, "n")	1	1s			
	2	2s	2p		
	3	3s	3p	3d	
	4	4s	4p	4d	4f
	5	5s	5p	5d	5f
	6	6s	6p	6d	6f
	7	7s	7p	7d	7f

Order: 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s 5f 6d 7p



Atomic number	Symbol	Electron configuration	Atomic number	Symbol	Electron configuration	Atomic number	Symbol	Electron configuration
1	H	1s ¹	37	Rb	[Kr]5s ¹	73	Ta	[Xe]6s ² 4f ¹⁴ 5d ³
2	He	1s ²	38	Sr	[Kr]5s ²	74	W	[Xe]6s ² 4f ¹⁴ 5d ⁴
3	Li	[He]2s ¹	39	Y	[Kr]5s ² 4d ¹	75	Re	[Xe]6s ² 4f ¹⁴ 5d ⁵
4	Be	[He]2s ²	40	Zr	[Kr]5s ² 4d ²	76	Os	[Xe]6s ² 4f ¹⁴ 5d ⁶
5	B	[He]2s ² 2p ¹	41	Nb	[Kr]5s ¹ 4d ⁴	77	Ir	[Xe]6s ² 4f ¹⁴ 5d ⁷
6	C	[He]2s ² 2p ²	42	Mo	[Kr]5s ¹ 4d ⁵	78	Pt	[Xe]6s ¹ 4f ¹⁴ 5d ⁹
7	N	[He]2s ² 2p ³	43	Tc	[Kr]5s ² 4d ⁵	79	Au	[Xe]6s ¹ 4f ¹⁴ 5d ¹⁰
8	O	[He]2s ² 2p ⁴	44	Ru	[Kr]5s ¹ 4d ⁷	80	Hg	[Xe]6s ² 4f ¹⁴ 5d ¹⁰
9	F	[He]2s ² 2p ⁵	45	Rh	[Kr]5s ¹ 4d ⁸	81	Tl	[Xe]6s ² 4f ¹⁴ 5d ¹⁰ 6p ¹
10	Ne	[He]2s ² 2p ⁶	46	Pd	[Kr]4d ¹⁰	82	Pb	[Xe]6s ² 4f ¹⁴ 5d ¹⁰ 6p ²
11	Na	[Ne]3s ¹	47	Ag	[Kr]5s ¹ 4d ¹⁰	83	Bi	[Xe]6s ² 4f ¹⁴ 5d ¹⁰ 6p ³
12	Mg	[Ne]3s ²	48	Cd	[Kr]5s ² 4d ¹⁰	84	Po	[Xe]6s ² 4f ¹⁴ 5d ¹⁰ 6p ⁴
13	Al	[Ne]3s ² 3p ¹	49	In	[Kr]5s ¹ 4d ¹⁰ 5p ¹	85	At	[Xe]6s ² 4f ¹⁴ 5d ¹⁰ 6p ⁵
14	Si	[Ne]3s ² 3p ²	50	Sn	[Kr]5s ¹ 4d ¹⁰ 5p ²	86	Rn	[Xe]6s ² 4f ¹⁴ 5d ¹⁰ 6p ⁶
15	P	[Ne]3s ² 3p ³	51	Sb	[Kr]5s ¹ 4d ¹⁰ 5p ³	87	Fr	[Rn]7s ¹
16	S	[Ne]3s ² 3p ⁴	52	Te	[Kr]5s ¹ 4d ¹⁰ 5p ⁴	88	Ra	[Rn]7s ²
17	Cl	[Ne]3s ² 3p ⁵	53	I	[Kr]5s ¹ 4d ¹⁰ 5p ⁵	89	Ac	[Rn]7s ² 6d ¹
18	Ar	[Ne]3s ² 3p ⁶	54	Xe	[Kr]5s ¹ 4d ¹⁰ 5p ⁶	90	Th	[Rn]7s ² 6d ²
19	K	[Ar]4s ¹	55	Cs	[Xe]6s ¹	91	Pa	[Rn]7s ² 5f ² 6d ¹
20	Ca	[Ar]4s ²	56	Ba	[Xe]6s ²	92	U	[Rn]7s ² 5f ³ 6d ¹
21	Sc	[Ar]4s ² 3d ¹	57	La	[Xe]6s ² 5d ¹	93	Np	[Rn]7s ² 5f ⁴ 6d ¹
22	Ti	[Ar]4s ² 3d ²	58	Ce	[Xe]6s ² 4f ¹ 5d ¹	94	Pu	[Rn]7s ² 5f ⁶
23	V	[Ar]4s ² 3d ³	59	Pr	[Xe]6s ² 4f ³	95	Am	[Rn]7s ² 5f ⁷
24	Cr	[Ar]4s ² 3d ⁵	60	Nd	[Xe]6s ² 4f ⁴	96	Cm	[Rn]7s ² 5f ⁷ 6d ¹
25	Mn	[Ar]4s ² 3d ⁵	61	Pm	[Xe]6s ² 4f ⁵	97	Bk	[Rn]7s ² 5f ⁹
26	Fe	[Ar]4s ² 3d ⁶	62	Sm	[Xe]6s ² 4f ⁶	98	Cf	[Rn]7s ² 5f ¹⁰
27	Co	[Ar]4s ² 3d ⁷	63	Eu	[Xe]6s ² 4f ⁷	99	Es	[Rn]7s ² 5f ¹¹
28	Ni	[Ar]4s ² 3d ⁸	64	Gd	[Xe]6s ² 4f ⁷ 5d ¹	100	Fm	[Rn]7s ² 5f ¹²
29	Cu	[Ar]4s ² 3d ¹⁰	65	Tb	[Xe]6s ² 4f ⁹	101	Md	[Rn]7s ² 5f ¹³
30	Zn	[Ar]4s ² 3d ¹⁰	66	Dy	[Xe]6s ² 4f ¹⁰	102	No	[Rn]7s ² 5f ¹⁴
31	Ga	[Ar]4s ² 3d ¹⁰ 4p ¹	67	Ho	[Xe]6s ² 4f ¹¹	103	Lr	[Rn]7s ² 5f ¹⁴ 6d ¹
32	Ge	[Ar]4s ² 3d ¹⁰ 4p ²	68	Er	[Xe]6s ² 4f ¹²	104	Rf	[Rn]7s ² 5f ¹⁴ 6d ²
33	As	[Ar]4s ² 3d ¹⁰ 4p ³	69	Tm	[Xe]6s ² 4f ¹³	105	Db	[Rn]7s ² 5f ¹⁴ 6d ³
34	Se	[Ar]4s ² 3d ¹⁰ 4p ⁴	70	Yb	[Xe]6s ² 4f ¹⁴	106	Sg	[Rn]7s ² 5f ¹⁴ 6d ⁴
35	Br	[Ar]4s ² 3d ¹⁰ 4p ⁵	71	Lu	[Xe]6s ² 4f ¹⁴ 5d ¹	107	Bh	[Rn]7s ² 5f ¹⁴ 6d ⁵
36	Kr	[Ar]4s ² 3d ¹⁰ 4p ⁶	72	Hf	[Xe]6s ² 4f ¹⁴ 5d ²	108	Hs	[Rn]7s ² 5f ¹⁴ 6d ⁶
						109	Mt	[Rn]7s ² 5f ¹⁴ 6d ⁷
						110	Ds	[Rn]7s ¹ 5f ¹⁴ 6d ⁹
						111	Rg	[Rn]7s ¹ 5f ¹⁴ 6d ¹⁰

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Periodic Table and Periodic Properties of Elements

- All elements are arranged on the periodic table.
 - The modern form of the periodic table was put forward by Mendeleev, a Russian chemist.
- The arrangement is based on the periodicity of the electronic structure of the atoms.
 - Such a pattern has been made possible because elements in the table have been arranged according to their atomic numbers.
- The elements of the periodic table can be divided into **groups, periods and blocks**.



Groups of a periodic table

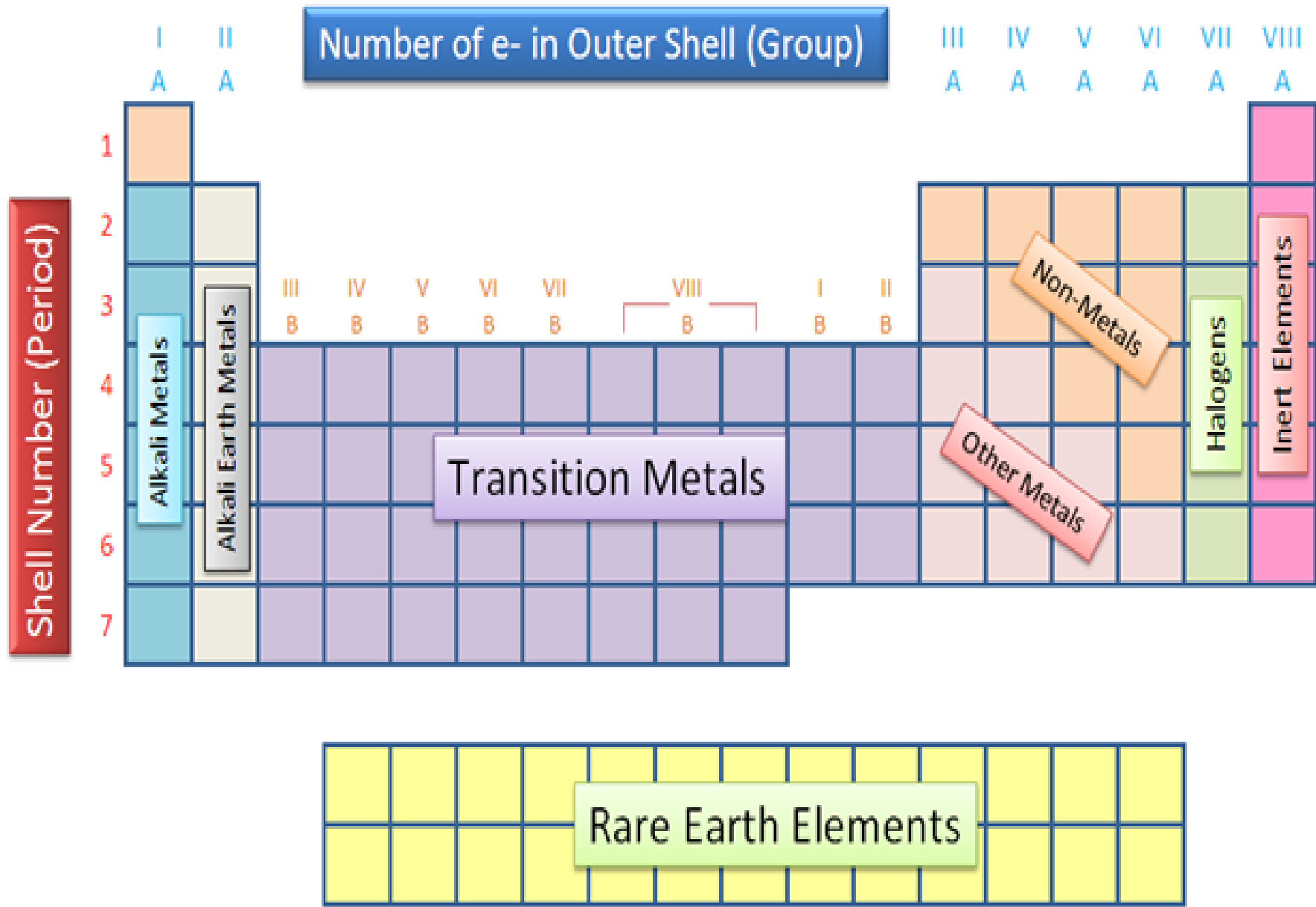
- A group is a vertical arrangement on the periodic table.
-
- The elements in a group have the same valence electrons or they have similar electronic configuration.
-
- Consider the electronic configuration of the elements; Li, Na, K, Rb, Cs and Fr.

Atomic number	Symbol	Electronic configuration
3	Li	$1s^2 2s^1$ (or) [He] $2s^1$
11	Na	$1s^2 2s^2 2p^6 3s^1$ (or) [Ne] $3s^1$
19	K	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ (or) [Ar] $4s^1$
37	Rb	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 5s^1$ (or) [Kr] $5s^1$
55	Cs	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^6 6s^1$ (or) [Xe] $6s^1$
87	Fr	[Rn] $7s^1$



- Where group members share similar physical and chemical properties, they could be considered as a 'family'.
 - For example, the alkali metals made up of Li, Na, K, Rb, Cs as well as the halogens or group VII elements form families.
- Where similarity of elements in the same group (congeners) is not marked, the tag 'family' should be avoided.
 - An example is group IV made up of C, Si, Ge, Sn and Pb.
- Elements in the same group have different n values.





Periods

- A period is a row or horizontal arrangement of elements on the periodic table.
- In a period, from one element to the next, the atomic number increases by one.
- In all, there are seven periods.
 - The length of the periods is not uniform, but varies.
 - The short period has two elements, and the longest should have thirty two elements.
 - The periods are numbered from 1 to 7.



- The period number refers to the main energy level that the valence electrons of the elements of a period occupy.
 - In period 1, the valence electrons of the elements occupy $n=1$.
 - There are only two elements in period 1; hydrogen and helium.
- The valence electrons of elements of the second period enter $n=2$ (or $2s$ $2p$).
 - The maximum number of electrons in $2s$ and $2p$ subshells is two and six respectively, giving a total of eight.
 - So the total number of elements in period 2 is eight.
- Next is the third period in which the valence electrons enter the main shell of principal quantum number, $n=3$. This has three orbitals $3s$, $3p$ and $3d$.
 - We would have expected the third period to have eighteen elements, but this period, like the second one, has eight elements.



- This is because the 3d-orbitals are not filled.
 - The reason for the non-filling of the 3d is that they have higher energy.
 - So the next electrons rather enter the 4s orbital, which is of lower energy than 3d.
 - However, 4s is of a higher quantum number, so it is considered a new shell, and the first element whose valence electron enters the 4s is not considered to be period 3 element
-
- Rather, a period 4 element in the fourth period, therefore, the orbitals that are filled by the valence electrons of the elements of the period are 4s, 3d and 4p, corresponding to eighteen elements.
-
- Similarly the valence electrons of the fifth period elements fill 5s, 4d and 5 p orbitals: this also gives eighteen elements



- The sixth period is made up of elements in which the valence electron fill 6s, 4f, 5d and 6p orbitals, giving the size of the period as 32 elements.
 - But then, accommodating such a large number of elements in one row would make the period unduly wide.
 - As a way of avoiding this clumsy arrangement, the elements with their valence electrons filling the 4f orbitals are placed in a separate row at the bottom of the periodic table.
 - This row of elements forms the lanthanide series. The lanthanides start from $_{58}\text{Ce}$ to $_{71}\text{Lu}$.
- In the seventh period, a similar modification is made whereby elements with their valence electrons filling 5f orbitals are placed below the lanthanides.
 - These are the Actinides; the actinides start from $_{90}\text{Th}$ to $_{103}\text{Lr}$.



It is worthy of note that a new period begins after a noble gas configuration with a duplet or octet.

Blocks of the periodic table

- The last subshell or orbital that contains the valence electrons is the s-orbital in both groups I and II elements.
- Therefore, these two groups of elements are also called s-block elements.
- We can also look at the electronic configuration of the second period elements from boron to Neon.
 - It is clear that the valence electrons of the above-listed second period elements are in s- and p-orbitals.
 - However, the last orbital occupied by the valence electrons is p- orbital.
 - It is for this reason that group IIIA to group O or VIIIB elements are called p-block elements.



- The d-block elements form the transition metals in which the valence electrons fill d- orbitals.
 - Such d-block elements in the fourth period (Sc-Zn) form the first transition series.
 - The d-block elements in the fifth period (Y-Cd) form the second transition series.
 - There is also the third transition series from Lu-Hg.
- Finally, there are the f-block elements: these are the lanthanides and actinides.
 - All the f-block elements in a period have the same outer electron configuration; either $6s^2$ or $7s^2$.
 - The properties of the lanthanides, for example are similar.
 - This is probably due to their having the same outer electronic configuration $6s^2$ (but the 4 f electrons differ).
 - The f-block elements are sometimes called the inner transition elements.



- Those elements with their valence electrons in s- and p-orbitals are referred to as A group of elements.
 - The group number refers to the number of electrons in the outermost orbital or orbitals.
- There are groups IA to VIIIA.
 - The group VIIIA is sometimes designated as group O.
 - The s-block and p-block elements are the representative or main-group elements.



1 H 1s																	2 He 1s
3 Li 2s	4 Be											5 B ←	6 C	7 N 2p	8 O	9 F	10 Ne →
11 Na 3s	12 Mg											13 Al ←	14 Si	15 P 3p	16 S	17 Cl	18 Ar →
19 K 4s	20 Ca	21 Sc ←	22 Ti	23 V	24 Cr	25 Mn 3d	26 Fe	27 Co	28 Ni	29 Cu	30 Zn →	31 Ga ←	32 Ge	33 As 4p	34 Se	35 Br	36 Kr →
37 Rb 5s	38 Sr	39 Y ←	40 Zr	41 Nb	42 Mo	43 Tc 4d	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd →	49 In ←	50 Sn	51 Sb 5p	52 Te	53 I	54 Xe →
55 Cs 6s	56 Ba	57 La ←	72 Hf	73 Ta	74 W	75 Re 5d	76 Os	77 Ir	78 Pt	79 Au	80 Hg →	81 Tl ←	82 Pb	83 Bi 6p	84 Po	85 At	86 Rn →
87 Fr 7s	88 Ra	89 Ac ←	104 Rf	105 Db	106 Sg	107 Bh 6d	108 Hs	109 Mt	110	111	112	113	114				
			58 Ce ←	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd 4f	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu →	
			90 Th ←	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm 5f	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr →	

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Periodic properties

- The physical and chemical properties of atoms which do recur on the periodic table are called periodic properties.
- The recurrence of these properties is due to some common features of the electronic structures.
- Some of the atomic properties which do recur are
 - Atomic / ionic radius
 - ionization energy
 - electron affinity
 - electronegativity



- In order to follow how these properties change on the periodic table, some terms have to be introduced and explained.
- These include
 - *Valence electrons*: Valence electrons or the outermost electrons are those electrons that are involved in bond formation. They have high energy.
 - *Core electrons*: The core electrons are the innermost electrons. They are between the nucleus and the valence electrons. Unlike the valence electrons, they are not involved in bond formation.
 - *Effective nuclear charge*: The effective nuclear charge refers to the net attraction of the nucleus for the outermost or valence electrons.



- ***Shielding/screening effect:*** The screening effect is the reduced attraction of the protons for the valence electrons due to the core electrons. These core electrons, using their negative charge, reduce the full effect of the nuclear attraction for the valence electrons. The core electrons, thus shield or screen the valence electrons from the full attraction of the protons in the nucleus.
- Across a period, electrons enter the same shell
 - Screening amongst electrons in the same shell is not much so effective nuclear charge increases.
 - On the other hand, as a group is descended new shells are added, the screening effect increases, and the effective nuclear charge virtually does not change even though the number of protons increases as the group is descended.



Atomic radius

- For covalently bonded atoms in a molecule, the atomic radius can be taken as half the distance between the two bonded nuclei.
- Metallic radius is also half the distance between the two nuclei of adjacent metal atoms.
- The size of an atom is dependent on how strongly the protons in the nucleus attract the valence electrons.
 - Across a period, electrons enter the same shell.
 - The screening effect due to electrons in the same shell is not much, almost constant.
 - However, the positive charge on the nucleus increases by one proton from one element to the next in a period.



- As a result, there is greater attraction for the valence electrons, and leads to a decrease in the size of the atom.
- But from the halogens to the rare gases, there is an increase.
 - This is attributed to the fact that the repulsion amongst electrons in a complete outer shell is marked.



Ionic radius

- Ions are charged atoms which are formed either by the loss or gain of electrons.
- There are two types of ions;
 - Cations
 - Anions.
- A cation is formed when an element loses an electron.
 - For example Li and Na can each lose an electron to form Li^+ and Na^+ respectively.
 - Li^+ , now has the electronic configuration of its preceding noble gas, He, while Na^+ has the electronic configuration of Ne.
- They are said to be isoelectronic with the noble gases.

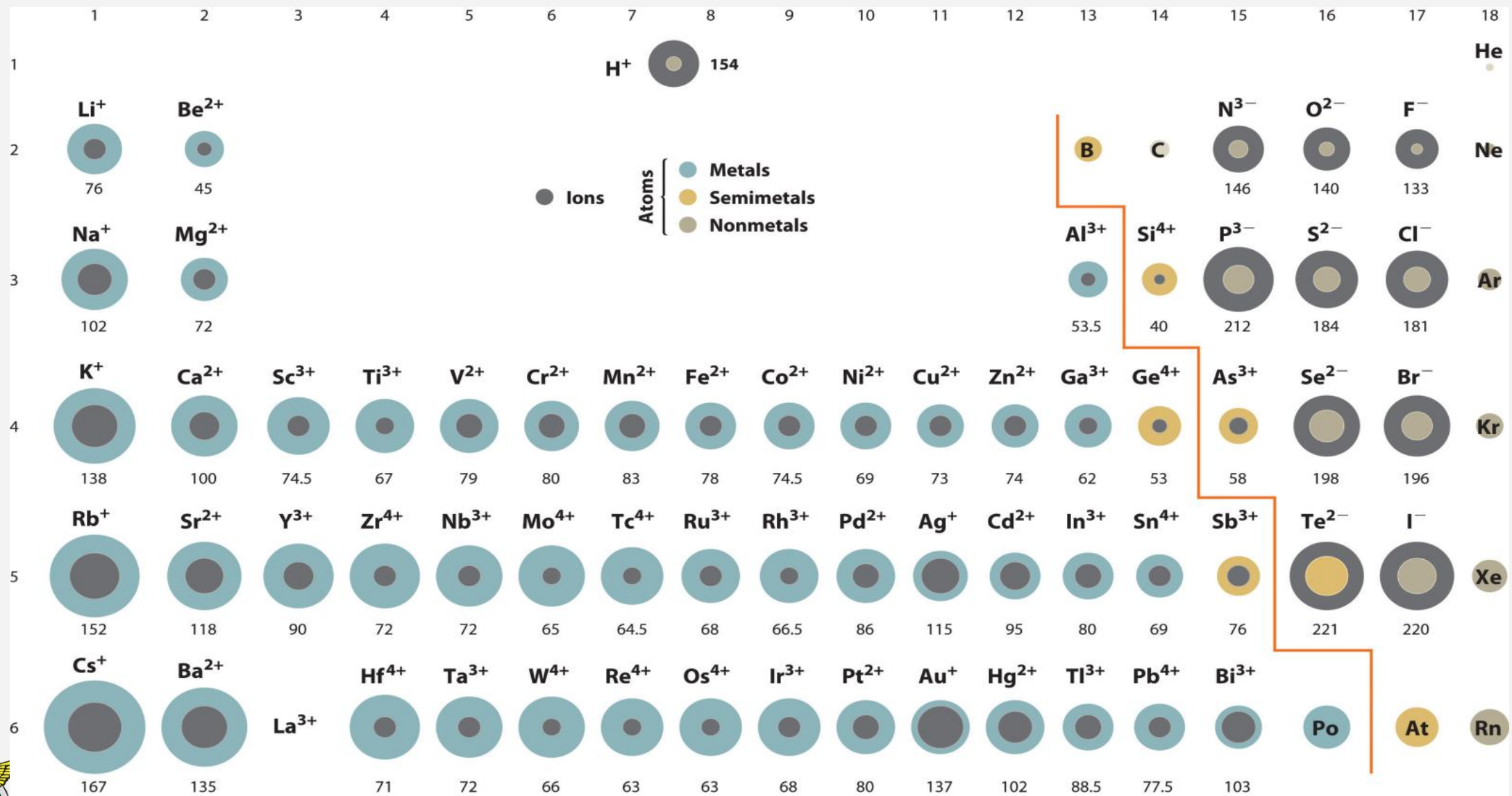


- A cation is smaller than its parent atom.
 - A cation has fewer number of electrons than the neutral atom.
 - So the fewer electrons are attracted more strongly, hence the size decreases.
 - The more the number of electrons lost, the greater the attraction and hence the smaller the size of the cation.
- We expect Mg^{2+} to be smaller in size than Na^+ due to the greater nuclear attraction in Mg^{2+} ion with a double positive charge.



- Anions are formed when atoms gain electrons.
 - An anion is isoelectronic with the noble gas which is immediately ahead of its atom.
- Anions are bigger than their parent atoms for two reasons.
 - Firstly, in the anions there are more electrons than protons, so the attraction for the outer electronic shells is less.
 - Secondly, the gaining of electrons increases the repulsive forces amongst the electrons.
- Just like atomic radius, cationic radius decreases across a period, but down a group, there is an increase.
 - In a series of ions with the same number of electrons (isoelectronic series) cationic radius decreases with increase in atomic number.
 - The ionic radius of the following isoelectronic series will be in the order;
 $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$.





- The more the number of electrons gained, the larger the size of the anion.
- Of the three ions, Al^{3+} is the smallest. Some values of atomic and ionic radius (in nm) are shown below

Atom	Li	Be	B	C	N	O	F	Ne
Atomic radius	156	112	85	77	71	60	69	131
Ionic radius	90	59	41	260	132	126	119	



Ionisation energy

- This is the energy required to expel the most loosely bound electron from a gaseous atom.
- Alternatively, it could be defined as the energy required to remove an electron from a gaseous atom to form one mole of gaseous cation with a single charge.
- The following factors determine ionization energy;
 - atomic radius
 - shielding effect
 - stability of electronic configuration
 - penetration effects



- As a group is descended, the increasing size of the atom, coupled with the screening effect make the attraction of the nucleus for the outermost electrons less.
 - Therefore, less energy is required to expel the outermost electrons.
- Across a period, there is a general increase in the ionization energy.
 - The effective nuclear charge increases, and atomic radius decreases.
 - Attraction for the outermost electrons increases, accounting for the increase in the ionization energy.
- However, there are some discrepancies in this trend across the period.
 - In the second period for instance, the ionization energy of Be ($1s^2 2s^2$) is higher than that of B ($1s^2 2s^2 2p^1$) while that N is higher than O.
 - The 2p electron of B is of higher energy than 2s electron of Be.



- The higher energy implies less stability, making the loss of an electron from 2p easier, and thus requiring less energy.
- It could also be said that the *s* electron has greater penetrating power than the 2p electron
- This makes the loss of a p-electron easier.
- It is because the *s* electron may be found closer to the nucleus than the p-electron that the former is said to have greater penetrating power (a p-electron will have greater penetrating effect than a d-electron).



- Another explanation that could be given is that there is greater stability associated with an outermost orbital which is fully filled as found in Be.
- One other irregular feature in the trend of ionization energies across the second period is the higher ionization energy of nitrogen ($1s^2 2s^2 2p^3$) as against that of oxygen ($1s^2 2s^2 2p^4$).
- In N, the 2p orbitals are half-filled, and there is stability associated with the half-filled orbitals as the single electrons in the p-orbitals are spread out, minimizing repulsion.

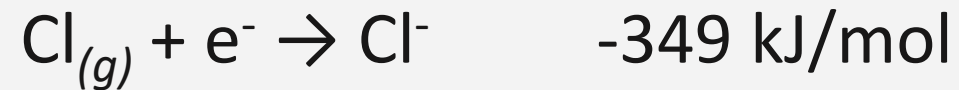


- Alternatively, we could ascribe the lower ionization energy of oxygen to its instability due to repulsion between the two electrons which become paired up in a p-orbital.
- To remove the second, third. . . . electrons in an atom (ion) requires the input of second, third... .etc.
 - Ionization energies which will be increasing.
 - The net positive charge makes it increasingly difficult to lose electrons in the successive steps.
 - Fewer electrons are being attracted by the same number of protons: in addition, the remaining electrons are more close to the nucleus.
 - Where a noble gas electronic configuration is involved, an unusually high ionization energy is required owing to the stability of the noble gas configuration.



Electron affinity

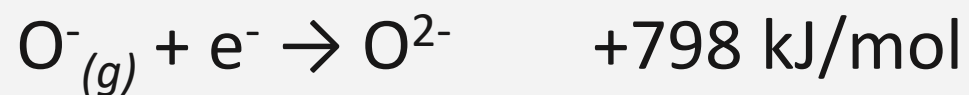
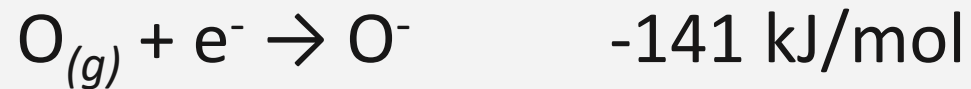
- This is the energy change when an electron is added to a gaseous atom.
- For a halogen, we could define electron affinity as the energy released when an electron is added to a gaseous atom to form a gaseous anion.



- The halogens have highly negative electron affinities.
- The electron which is being added faces repulsion from the shell it has filled, but there is also the attraction between the added electron and the protons



- It is this attraction that accounts for the release of energy.
- A caution should be taken here: it is not in all cases that electron affinity is a negative energy value.
- Take the case of oxygen which can step wisely gain two electrons.



- The second electron is added against the repulsive force of the negatively charged oxygen.
- The repulsive force must be overcome by an energy input, hence the positive value of electron affinity.



- The way the electron affinity varies is more complex than the variation of ionization energy.
- All the same, electron affinity depends on
 - atomic radius
 - stability of electronic configuration
 - for anions, the charge carried on them
- Down a group the electron affinity decreases because atomic radius increases and attraction for the added electron gets less.
- Across the same period, electron affinity increases, effective nuclear charge increases, resulting in the increased attraction for the added electron.
- The incoming electron is attracted more strongly by the decreasing atomic size as well as the increasing positive charge on the nucleus.

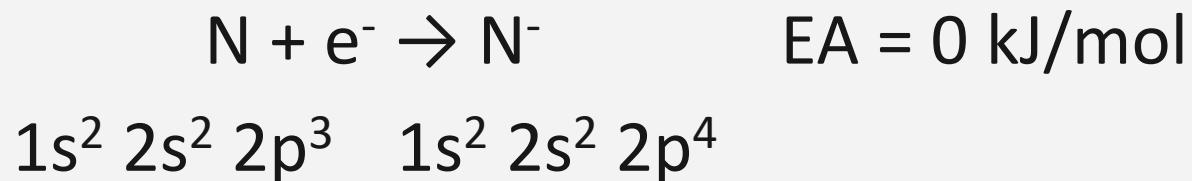
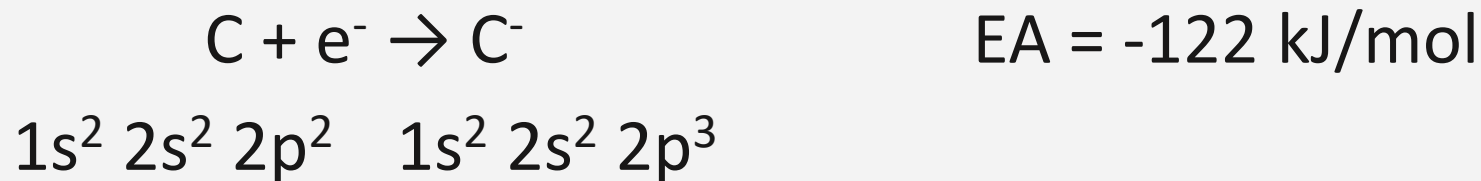
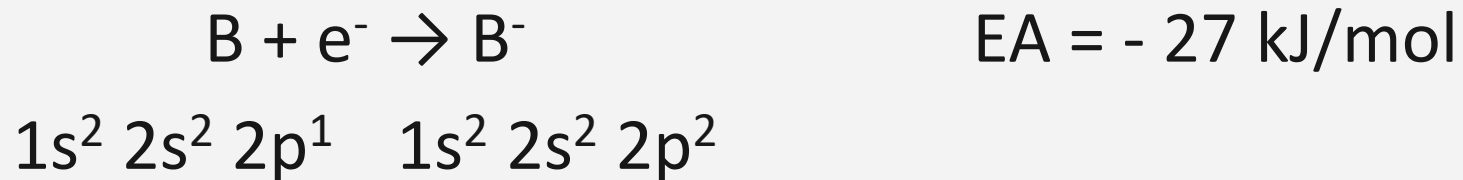


Electron Affinities (kJ/mol)

1A							8A
H -73							He >0
	2A	3A	4A	5A	6A	7A	
Li -60	Be >0	B -27	C -122	N >0	O -141	F -328	Ne >0
Na -53	Mg >0	Al -43	Si -134	P -72	S -200	Cl -349	Ar >0
K -48	Ca -2	Ga -30	Ge -119	As -78	Se -195	Br -325	Kr >0
Rb -47	Sr -5	In -30	Sn -107	Sb -103	Te -190	I -295	Xe >0



- The electron affinities of B, C and N are as shown below



Electronegativity

- Electronegativity is the ability of an atom in a molecule to attract shared electrons.
- A distinction should be made between electronegativity and electron affinity.
 - In defining electron affinity, we consider gaseous isolated atoms whereas in electronegativity we consider atoms in molecules.
- Electron affinity is an energy input or output, having units of kJ/mol; it could be positive or negative.
- Electronegativity is a comparative scale, giving a quantitative measure of atom's ability to attract electrons.
 - It has no units or it is a dimensionless number. Its values are always positive.



- Values of electronegativity are given according to the Pauling's scale.
- The highest value of 4.0 is given to fluorine in group VIIA while the lowest value of 0.7 is assigned to Francium in group IA.
- Another difference between electron affinity and electronegativity is that the former can be directly measured whereas the latter is measured indirectly.
- Like electron affinity, electronegativity increases across a period but decreases down a group.
- The most electronegative atoms are the reactive non- metals, the halogens which are on the right side of the periodic table.



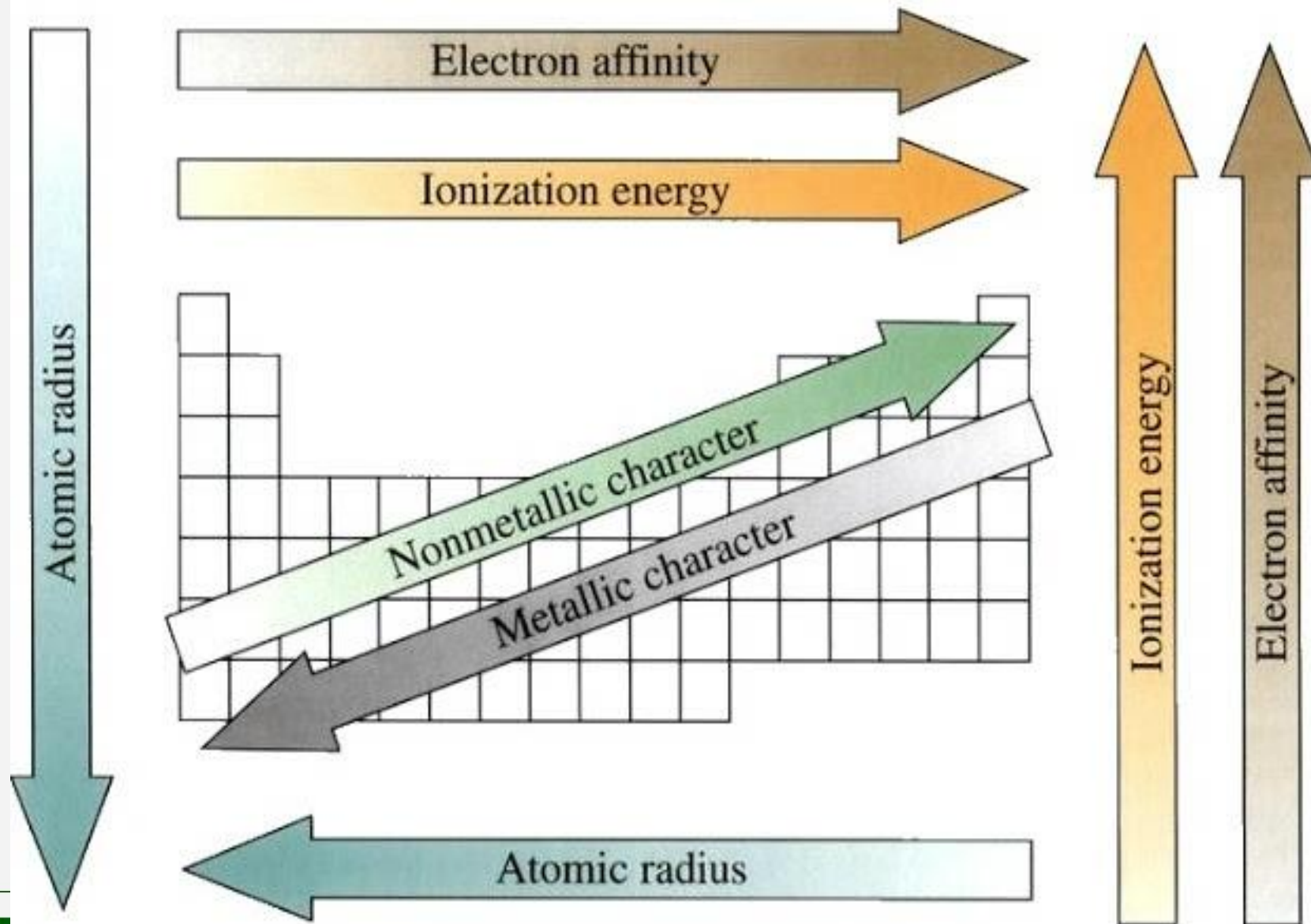
- On the other hand, the least electronegative atoms are the reactive metals, group IA on the left of the periodic table.
- We will come to see later that the electronegativity difference between atoms in compounds can be used to predict the type of bonding in compounds.



H 2.20																	He n.a.
Li 0.98	Be 1.57											B 2.04	C 2.55	N 3.04	O 3.44	F 3.98	Ne n.a.
Na 0.93	Mg 1.31											Al 1.61	Si 1.90	P 2.19	S 2.58	Cl 3.16	Ar n.a.
K 0.82	Ca 1.00	Sc 1.36	Ti 1.54	V 1.63	Cr 1.66	Mn 1.55	Fe 1.83	Co 1.88	Ni 1.91	Cu 1.90	Zn 1.65	Ga 1.81	Ge 2.01	As 2.18	Se 2.55	Br 2.96	Kr 3.00
Rb 0.82	Sr 0.95	Y 1.22	Zr 1.33	Nb 1.60	Mo 2.16	Tc 1.90	Ru 2.20	Rh 2.28	Pd 2.20	Ag 1.93	Cd 1.69	In 1.78	Sn 1.96	Sb 2.05	Te 2.10	I 2.66	Xe 2.60
Cs 0.79	Ba 0.89	La 1.10	Hf 1.30	Ta 1.50	W 2.36	Re 1.90	Os 2.20	Ir 2.20	Pt 2.28	Au 2.54	Hg 2.00	Tl 1.62	Pb 2.33	Bi 2.02	Po 2.00	At 2.20	Rn n.a.
Fr 0.70	Ra 0.89	Ac 1.10	Rf n.a.	Db n.a.	Sg n.a.	Bh n.a.	Hs n.a.	Mt n.a.	Ds n.a.	Rg n.a.	Uub n.a.	—	Uuq n.a.	—	—	—	—



Summary of periodic properties



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Rate of reactions

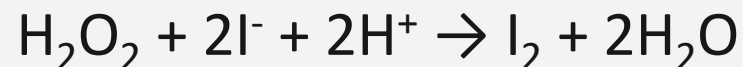
- During chemical reactions, reactants are converted to products.
- Consider the reaction which A is converted to B; $A \rightarrow B$.
- The reaction rate could be taken as either the rate at which the product (B) is formed or the rate at which reactant (A) disappears.

- $$\text{Rate of Reaction} = \frac{\text{Change in Quantity}}{\text{Time}}$$

- $$\text{Rate of Reaction} = \frac{\text{Change in Concentration}}{\text{Time}}$$



- On adding acidified H_2O_2 to iodide solution, the following reaction occurs:



- The concentration of iodine formed in a specified time can be determined by titration with sodium thiosulphate, $\text{Na}_2\text{S}_2\text{O}_3$, using starch as the indicator.
- Assuming that the concentration of iodine formed in 30 seconds was 0.30 mol/dm^3 , then the rate of formation of iodine can be worked out as

$$\begin{aligned}\text{Rate of Iodine Formation} &= \frac{\text{Change in Conc}}{\text{Time}} \\ &= \frac{0.30}{30} \text{ mol/dm}^3 \text{ sec} \\ &= 0.01 \text{ mol/dm}^3 \text{ sec}\end{aligned}$$



Ways of following the rate of reaction

- We could follow the rate of formation of products or the rate of disappearance of reactants.
- In the determination, we have to depend on some property of the reactants or products which changes as the reaction progresses.
- At some suitable time intervals, the reaction mixture is analysed in order to determine the concentration of products or the remaining reactants.
- The method used for determining the concentration depends on the nature of reactants and products.



Measurement of colour intensity

- One property which can be easily followed is colour changes. The reactant could be coloured, but is converted to a colourless or less coloured product.
- In such a case, we could follow the rate at which the colour intensity decreases.
- Or it could be a colourless reactant giving rise to a coloured product; so there will be an increase in colour intensity.
- It is also possible that neither the remaining reactant nor the formed product is coloured by itself.

But this may be made to undergo some chemical reaction to produce a coloured product which can then be measured.



- The colour intensity is measured with an instrument called the ***colorimeter***, or the spectral properties of the reaction mixture are measured using the spectrophotometer.

- Consider the reaction below:

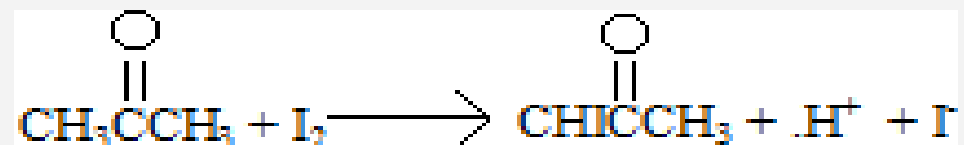


- Bromine is reddish brown, but none of the products is coloured.
- Therefore, there is decrease in colour intensity as the reaction proceeds.



Finding rate by titration

- Consider the reaction between propanone and iodine.



- At timed intervals, a small volume of the reaction mixture is taken.
- It has to be ensured that the reaction in the aliquot sample is stopped immediately;
 - the reaction is quenched by adding a reagent which prevents further reaction from taking place.
 - In this example in which the acid serves as a catalyst, NaHCO_3 could be added to neutralize the acid.



- Another means of quenching is to put the sampled mixture on ice to lower the temperature in order to drastically reduce the rate of reaction or to stop the reaction altogether.
- The sampled mixture can be titrated against $\text{Na}_2\text{S}_2\text{O}_3$ using starch as the indicator.



Use of pressure measurements for reactions involving gases

- In such reactions, there will be changes in pressure in the reaction vessel at constant volume. For example,



- There will be an increase in pressure because of the evolution of hydrogen gas.

- For the reaction,



- there will be a decrease in pressure because 4 moles of gas yield three moles of gas.



Conductivity measurements

- Conductivity is the measure of ions in solution.
- In some reactions, ions are either produced or consumed, so the increase or decrease in number of ions can be monitored using the conductivity meter.

Viscosity measurement

- Where a reaction involves changes in viscosity, this could be followed using the *viscometer*.
- For example, the rate of flow of a liquid in a narrow tube can be used to measure viscosity.
- The rate of hydrolysis of starch by acid or enzyme can be followed this way.



Rotation of plane polarized light using a polarimeter

- When sucrose is hydrolysed by an acid or enzyme to form an invert sugar, the rate of inversion can be monitored by measuring the change in optical rotation of the sucrose solution in given time intervals.

Measurement of radioactivity

- In radiochemical assay a reactant is radioactively labelled, leading to the formation of a product which would be radioactive.
- When the reaction is stopped in a certain time the product is separated from the reactant, and the amount of radioactivity in the product provides a measure of the rate.
- The rate of incorporation of Fe in haemoglobin in mammals can be determined this way.



Reaction rate and stoichiometry

- For a reaction of simple stoichiometry like $A \rightarrow B$, we can express the rate of reaction by measuring the rate in terms of the change in the concentration of either reactant or product.
- $Rate = -\frac{\Delta[A]}{\Delta t}$ or $Rate = \frac{\Delta[B]}{\Delta t}$
- The rate of formation of the products does not require the negative sign as $\Delta[B]$ is a positive quantity.
- For more complex reactions, the coefficients of the reactants and products should be taken care of in the derived equation.
- Consider the reaction $2A \rightarrow B$



- According to the stoichiometry, two moles of A disappear as one mole of B is formed.
- The rate of disappearance of A is two times as fast as B.
- Therefore the rate could be written either
- $Rate = -\frac{\frac{1}{2}\Delta[A]}{\Delta t}$ or $Rate = \frac{\Delta[B]}{\Delta t}$
- Generally for the reaction of the type $aA + bB \rightarrow cC + dD$, the rate relation is as follows.
- **Problem 1**
- For the reaction $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$, write the rate expression in terms of the disappearance of reactants and the appearance of products.



Factors affecting rates of reaction

- Nature of reactants
- Concentration of reactants
- Temperature
- Presence of catalysts
- Pressure (for gaseous reactants)



Nature of reactants

- Reactions involving different reactants occur at widely different rates even though the conditions may be the same.
- For example, neutralization reaction takes place in microseconds at even room temperature.
- On the other hand, esterification will be slow at room temperature.
- An acid catalyst has to be used; in addition heat has to be applied.
- Geological reactions like formation of rocks take millions of years.
- The difference in rates is accounted for by the different nature of reactants which have different activation energies.



- Organic reactions are generally slower than reactions involving inorganic compounds.
- For inorganic compounds in aqueous solution, the ions are already present and they tend to combine in a fast reaction.
- But organic compounds are held by covalent bonds, and it is in the course of the reaction that charged intermediates are formed to facilitate many of such organic reactions.
- Therefore higher activation energies are involved in organic reactions.
- Furthermore, in organic reactions, bond strength will also determine the rate of reaction; the activation energy will be dependent on the bond strength.



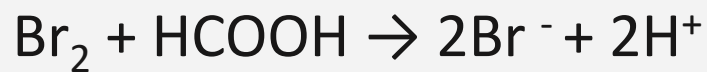
- The sizes and shapes of organic molecules will also affect the rate of reaction.
- Large molecules with bulky groups may mask the reactive part of the molecule by steric hindrance.
- An illustration is the observation that aldehydes react faster than ketones in nucleophilic reactions partly due to the presence of more bulky groups in the ketones.
- For reactions that occur in two phases (gas/solid or liquid/solid), the state of subdivision of particles will determine the rate.
- A lump of calcium carbonate will react with HCl while powdered form of CaCO_3 will react very fast.
- The reaction the carbonate and the acid takes place at the surface where the reactants come into contact with each other.



■ Therefore, the larger the surface area, the faster the reaction.

Concentration of reactants

- In most reactions, increasing the concentration of reactants would lead to an increase in the rate.
- However, there are some reactions in which rate is unaffected by changes in reactant concentration.
- In addition, the magnitude in rate changes as a function of changes in reactant concentration are not fixed, but vary as will be shown presently.
- Recall the reaction between bromine and methanoic acid;



- We have already been made aware that as the bromine is reduced by the methanoic acid, the reddish brown colour of bromine is discharged.
- So we can follow the rate reaction by measuring the decrease in colour intensity of bromine with time.
- Table 1

Time (secs.)	Bromine concentration (M)	Average rate (mol/sec)
0.0	0.0120	
50	0.0101	3.8×10^{-5}
100	0.00846	3.28×10^{-5}
150	0.00710	2.72×10^{-5}
200	0.00596	2.28×10^{-5}
250	0.00500	1.92×10^{-5}
300	0.00420	1.60×10^{-5}



- From the data given, the average rate of reaction can be calculated.
- The average rate is defined as the change in bromine concentration over a specified time.

$$\text{Average Rate} = \frac{[Br_2]_{final} - [Br_2]_{initial}}{T_{final} - T_{initial}} = - \frac{\Delta[Br_2]}{\Delta t}$$

- Since the bromine concentration decreases with time, the change in bromine concentration, $\Delta[Br_2]$ is a negative value.
 - However, rate of reaction should be a positive quantity.
- Therefore, a minus sign is introduced in the rate expression to make the rate a positive value.



- From the table 1 the average rate in the first 50 seconds is 3.8×10^{-5} mole/sec.

- This can be calculated as follows:

$$\begin{aligned}\text{Average Rate} &= - \frac{\Delta [\text{Br}_2]}{\Delta t} = \frac{[\text{Br}_2]_{\text{final}} - [\text{Br}_2]_{\text{initial}}}{T_{\text{final}} - T_{\text{initial}}} \\ &= \frac{-0.0101 - 0.0120}{50 - 0} \\ &= \frac{0.0019}{50} \\ &= 3.80 \times 10^{-5} \text{ mole/sec}\end{aligned}$$

- The average rate in the next 50 seconds will be as follows

$$\begin{aligned}&= \frac{-0.00846 - 0.0101}{100 - 50} \\ &= 3.28 \times 10^{-5} \text{ mole/sec}\end{aligned}$$



- From the two calculations carried out, as well as from Table 1 the average rates of reactions are not constant, but change with the concentration of the reactant molecules.
- Initially, when the concentration of bromine is high, the rate is also high.
- But with time, the bromine concentration gradually decreases until it eventually becomes zero when all the bromine has been used up.
- Even though the reaction rate also depends on the concentration of methanoic acid, if the acid concentration is maintained at a high level, it can be assumed that the acid concentration is almost constant.



- Using average rates to describe the progress of a reaction is not informative enough as the average rates calculated do not correspond to specific times.
- For example the average of 3.80×10^{-6} mole/sec stands for the rate from $t = 0$ to $t = 50$ secs.



Instantaneous rates

- In the determination of instantaneous rates, a graphical approach is used.
- Still using the bromine example, a graph of bromine concentration against time can be drawn as shown below

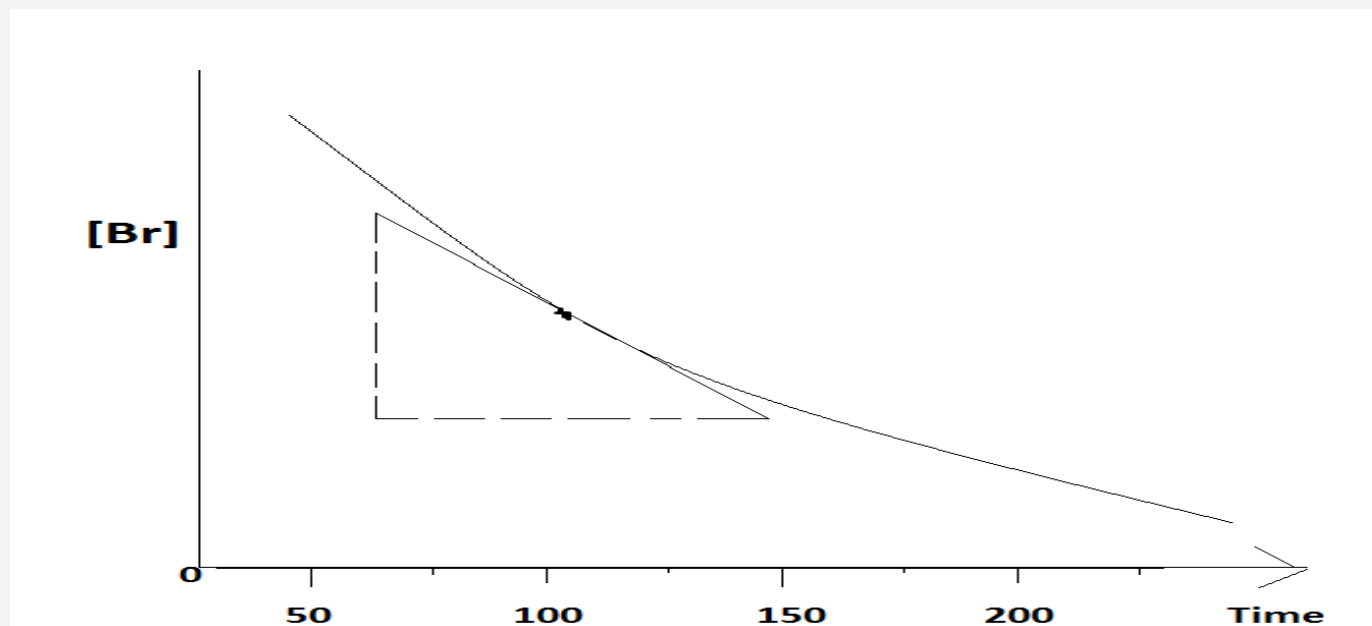


Fig 1 Graph of bromine concentration against time.



- To find the rate at any specific time (instantaneous rate), a tangent is drawn to the curve at that particular time.
- The gradient to that tangent gives the instantaneous rate.
- A tangent has been drawn to the curve at 50 secs.
- The gradient of this tangent will give the instantaneous rate at $t = 100$ secs.
- Other tangents can be drawn to the curve at 50, 150, 200 secs. Gradients to these tangents would give the various instantaneous rates.
- In fact, a tangent can even be drawn at zero time.



The gradient will be the instantaneous rate at zero time which should be the initial rate.

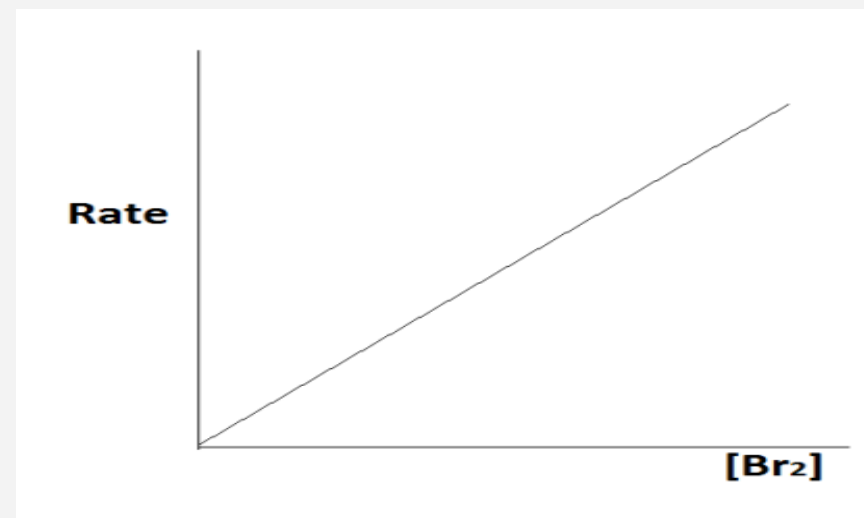
- Next, another graph can be drawn in which the bromine concentration is plotted against the instantaneous rates.

- If this gives a straight line, it means that the rate is proportional to the bromine concentration.

- $\text{Rate} \propto [\text{Br}_2]$

- Or $\text{rate} = k[\text{Br}_2]$ where k is the rate constant

- $k = \frac{\text{rate}}{[\text{Br}_2]} \text{ sec}^{-1}$



- k is not affected by concentration of bromine.
- But the rate is affected by the concentration of bromine: high concentration of bromine gives a high rate of reaction while low concentration will give a corresponding low rate.



The rate law

- The rate law or expression or equation relates the rate of reaction to the concentration of reactants.
- It is how the initial rate varies with reactant concentration which is considered.
- It is preferable to use the initial rate because as the reaction proceeds, the concentration of the reactants decreases, and it may become difficult to measure the changes accurately.
- Secondly, there may be a reverse reaction (product $\rightarrow$ reactant) which may introduce some error in the rate measurement.



- For a reaction of the type, $aA + bB \rightarrow cC + dD$: the rate law could be of the form

$$\text{Rate} = k [A]^x [B]^y$$

- Where k is the rate constant or specific rate constant, x is the order of reaction with respect to A, and y is the order of reaction with respect to B.
- The power to which a reactant concentration in the rate equation is raised is called the order.
- The sum of the orders of the various reactants in the rate law gives the overall order of reaction .
- In this example the overall order of reaction is $x + y$.
- The order and specific rate constant have to be determined experimentally .
- If a reaction involves only one reactant, the rate law can be determined by measuring the rate of reaction as a function of the concentration of the reactant.



- If the initial rate doubles when the concentration of reactant is doubled, then the reaction is first order with respect to the reactant.
- If the rate quadruples or doubly doubles when the concentration reactant is doubled, then the reaction is second order with respect to the reactant.
- If a reaction is made up of more than one reactant then the isolation technique can be used to determine the rate law.
- In this approach only one of the reactant concentration varied while the concentration of the other reactants is kept constant.
- The same procedure will be followed to find how the rate alters when the concentration of the other reactants varied.



Problem 1

- There is this reaction; $A + B \rightarrow C$
- The initial rates at the various initial reactant concentrations are given below.

Experiment	[A] M	[B] M	Initial rate (mole/sec)
I	0.1	0.1	4×10^{-4}
II	0.1	0.2	4×10^{-4}
III	0.2	0.1	16×10^{-4}

- From the provided information,
- i. Determine the rate law
- ii. What is the order of reaction with respect to A and B?
- iii. What is the overall order of reaction?
- iv. Calculate the rate constant for the reaction



Solution

- i. From experiment I and II, keeping the concentration of A constant, but doubling the concentration of B, the rate does not change. Therefore, rate $\propto [B]^0$.
- From experiment I and III, keeping the concentration of B constant, but doubling the concentration of A, the rate doubly doubles or quadruples. Therefore,

$$\text{Rate} \propto [A]^2$$

$$\text{Rate} = k [B]^0 [A]^2$$

$$\text{Or rate} = k [A]^2$$

- ii. The order of reaction with respect to A is 2, and the order of reaction with respect to B is zero.
- iii. The overall order of reaction is 2.



- iv. To calculate the rate constant, we use the rate law, $\text{rate} = k [A]^2$
- We can then use the data for any of the experiments to calculate the rate constant, k .
- For example we can use the data for experiment I
- $4 \times 10^{-5} = k (0.1)^2$

$$k = \frac{4 \times 10^{-5}}{(0.1)^2}$$

$$= 4.0 \times 10^{-3} \text{ mole}^{-1} \text{ sec}^{-1}$$



Note the following.

- The rate law is not usually determined from the stoichiometry of the equation
- The rate law could have the order being fractional
- The reaction rate could be zero order with respect to some reactants as shown in problem 1.
- In such a reaction the rate does not change when the reactant concentration is varied.
- $\text{Rate} = k[A]^0$ But $[A]^0 = 1$
- Therefore, $\text{rate} = k$



- An example of zero order reaction is the decomposition of gases on solid surfaces in which the rate does not depend on the concentration of reactant, but rather depends on the surface area of the solid.
- Another example of a reaction that follows zero order kinetics is when saturating concentration of substrate is used in enzymatic reactions.
- Under such a condition, the rate of the enzymatic reaction becomes independent of the substrate concentration.
- Sometimes, the reaction orders are the same as the coefficients in the equation.
- This holds if the reaction step is rate-determining.



- There is also what is called the pseudo-first order reactions.
- An example is hydrolytic reactions.
- When sucrose, for example, is hydrolysed the two reactants are sucrose and water:
 $\text{sucrose} + \text{water} \rightarrow \text{fructose} + \text{glucose}$
- Such a reaction should have been second order reaction in that the rate should be first order with respect to sucrose and water.
- But due to the high concentration of water, its concentration is assumed to be constant.
- Therefore, the rate appears to depend on only the sucrose concentration.



How to determine the order of reactions

- For a reaction of the type, $A \rightarrow B$, the order of reaction with respect to A can be found by measuring the concentrations of A at various times, t.
- Then a graph of [A] versus time can be plotted.
- Tangents can be drawn to the curve at various times, and the gradients of the tangents found.
- The gradients at the various times will give the rates,

$$-\frac{d[A]}{dt}$$

- For first order reactions, $-\frac{d[A]}{dt} = -k_1[A]$



- A graph of $\frac{d[A]}{dt}$ vrs $[A]$ gives a straight line with a gradient, k_1
- In the case of second order reactions, rate or $[A]^2$
- A plot of $\frac{d[A]}{dt}$ vrs $[A]$: gives a straight line with a gradient k_2
- Such an approach for obtaining the rate constant and order of reaction is tedious as we draw two graphs, draw tangents and then find gradients.



Integration of rate equation

- The integration of the rate law provides a more simplified way of finding the rate constant and order of reactions.
- Here, only one graph is drawn, and there is no need to draw tangents

Integration of equation for first order reactions

- Assume the reaction $A \rightarrow B$ is a first order reaction; then

- $$-\frac{d[A]}{dt} = k_1 [A]$$

- Rearrange

- $$\frac{d[A]}{[A]} = -k_1 dt$$



- Integrate between the limits $[A]_0$ at $t = 0$ and $[A]_t$ at time t .

- $$\int_{[A]_0}^{[A]_t} \frac{d[A]}{dt} = -k_1 \int_0^t dt$$

- $$\frac{\ln [A]_t}{[A]_0} = -k_1 t$$

- This can be rearranged to

- $$\ln [A]_t - \ln [A]_0 = -k_1 t$$

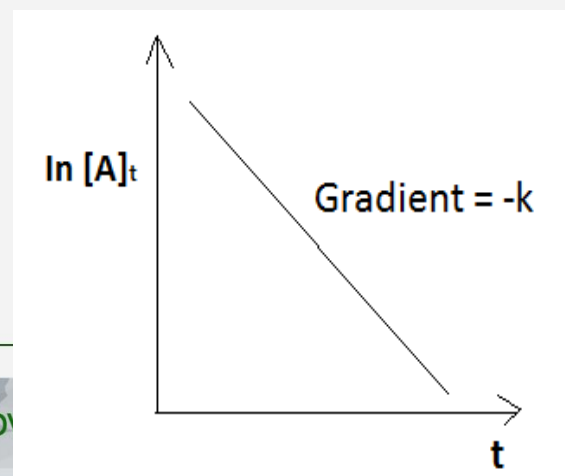
- $$\ln [A]_t = -k_1 t + \ln [A]_0$$

- The last equation can also be transformed to $\log_{10}$

- $$\log [A]_t = -\frac{kt}{2.3} + \log [A]_0$$



- The last equation can also be transformed to $\log_{10}$
- $\log [A]_t = -\frac{kt}{2.3} + \log [A]_0$
- The last two equations are in the form of an equation for a straight line; $y = mx + c$.
- A plot of $\ln[A]_t$ against t gives a straight line of slope $-k_1$.
- This allows the calculation of the rate constant for the first order reaction, k_1 .
- The intercept on the y-axis will be $\ln[A]_0$



Half-life of first order reactions

- From the integrated rate equation for first order reactions, we know that

$$\ln \frac{[A]_t}{[A]_0} = -k_1 t$$

- By inverting the equation we obtain

$$\ln \frac{[A]_0}{[A]_t} = k_1 t$$

$$t = \ln \frac{[A]_0}{[A]_t} \times \frac{1}{k}$$

- By the definition of half life, when $t = t_{1/2}$

$$[A]_t = \frac{[A]_0}{2}$$

$$t_{1/2} = \ln \frac{[A]_0}{[A]_0/2} \times \frac{1}{k}$$

$$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}$$



- The equation for a first order reaction shows that for such a reaction, the half life is constant and does not depend on the initial concentration.
- A typical example of a reaction that follows first order kinetics is radioactive decay. In this process the rate constant is called the decay constant, designated as λ .
- $t_{1/2} = \frac{0.693}{\lambda}$
- Consider ${}_{205}\text{U}$ which has its decay constant to be $4.80 \times 10^{-4}/\text{sec}$, its half life can be worked out as follows

$$t_{1/2} = \frac{0.693}{4.8 \times 10^{-4}/\text{sec}}$$

$$= 1444 \text{ secs.}$$
- For a second order reaction, $t_{1/2} = \frac{1}{k [A]_0}$



- In this case, the half-life depends on the initial concentration and increases with time.
- Thus, for second-order reactions, the half-life changes with time.

Problem 2

The half life of a first order reaction is 800 secs. If the initial concentration of the substance undergoing reaction is 0.8M,

- a) what will be the concentration after 1600 secs?
- b) how long will it take for the concentration to reduce to 0.1M?

Solution

- a) 1600 represents two half lives. In one half life the remaining substance will be 0.4M, and in another half life the concentration left will be 0.2M



$$t_{1/2} = \frac{0.693}{k}$$

$$800 = \frac{0.693}{k}$$

$$k = 8.662 \times 10^{-4} \text{ sec}^{-1}$$

$$\ln \frac{[A]_0}{[A]_t} = kt$$

$$\ln \frac{0.8}{0.1} = 8.662 \times 10^{-4} \times t$$

$$t = \frac{2.07944}{8.662 \times 10^{-4}}$$

$$= 2400 \text{ secs.}$$



- In this case where the time obtained is a whole number multiple of the half life, we could have arrived at the answer without using a formula.
- In this approach, you start with $t = 0$ and its corresponding concentration, and you can work through the 1st, 2nd, 3rd, etc half lives and their corresponding concentrations.

Time	Conc. In mole/l
0	0.80
800	0.40
1600	0.20
2400	0.1



- If we had been asked the time at which there will be left 0.3M of the sample, using the formula provides a better option.

$$\ln \frac{[A]_0}{[A]_t} = kt$$

$$\ln \frac{0.80}{0.30} = 8.662 \times 10^{-4} \times t$$

$$t = 1132 \text{ secs}$$

- The distribution of drugs in the body follows first order kinetics.



- From problem 2, we can see that when drugs are administered, we can use the integrated rate law to find the following:
 - The time required for an initial concentration to fall to a specified concentration
 - The concentration that will be attained in a given time.
 - The initial concentration, if the concentration after a certain period of time is given.
- The above calculations are important parameters in the therapeutic drug monitoring in patients to ascertain the pharmacological efficacy as well as any toxic side effects of drugs.
- The distribution and metabolism of insecticides also tend to allow first order kinetics.
- The half lives of the various insecticides would influence the efficacy of these insecticides in the control of pests.
- They would also determine the levels of pesticide residues in food chains.



Using radiocarbon dating for determining the age of old plant and animal parts.

- Radiocarbon dating was developed by an American physical chemist W.F Libby.
- He won a Nobel Prize for his work in 1960. Radiocarbon dating serves as a molecular clock to determine the age of old materials of plant and animal origin.
- Radiocarbon dating is based on the fact that there is simultaneous production and decay of MC.
- This radioactive carbon is produced as MN of the air captures a neutron according to the equation; $^{14}_7\text{N} + ^1_0\text{n} \rightarrow ^{14}_6\text{C} + ^1_1\text{H}$
- The neutrons are produced by high energy cosmic rays that pass through the atmosphere and strike atoms to strip off the neutrons.



- At the same time that ^{14}C is being formed, it decays by β -emission; $^{14}_6\text{C} \rightarrow ^{14}_7\text{N} + ^0_{-1}\beta$
- Due to the simultaneous production and decay of $^{14}_6\text{C}$, its level in the atmosphere remains constant.
- The ^{14}C of the atmosphere can form $^{14}_6\text{CO}_2$.
- There is also non-radioactive (C) in the atmosphere, also forming $^{14}_6\text{CO}_2$.
- Plants fix CO_2 in their bodies through photosynthesis.
- Animals feed on plants and thus get the radioactive and non-radioactive carbon incorporated in their tissues.



- On the death of the plant or animal, no more MC enters the body but the decay continues.
- For a living plant or animal, because of the constant intake of carbon, they are able to maintain a ratio of $^{14}\text{C} : ^{12}\text{C}$ that is identical to that of the atmosphere.
- But on the death of the organism, there is no longer any intake of the carbon to replenish the ^{14}C which is lost through the radioactive decay.
- Therefore, the ratio of $^{14}\text{C} : ^{12}\text{C}$ decreases.
- If the ratio diminishes to half that of the atmosphere, it implies the age of that object is one half life.



The half life of ^{14}C is about 5730 years.

- The following assumptions are made in ^{14}C dating.
- There is uniform distribution of ^{14}C over the earth.
- The steady state concentration of ^{14}C in the atmosphere observed today has been the same for several thousands of years. This assumption is incorrect.
- Radioactive decay rates are constant, not dependent on temperature, pressure and physical states.
- The short half life of ^{14}C is a limitation in radiocarbon dating.
- It cannot be used to determine the age of coal or petroleum deposits that have been laid down so long ago that no ^{14}C remains.



- It is of more value in establishing dates of more recent materials.
- It cannot be used for objects more than 25,000 years old.
- For materials older than this, the β -emission will be so weak that there will be interference by background radiation
- Two problems given below will illustrate the steps which are gone through radiocarbon dating.
- The ^{14}C content of the old plant or animal body is determined.
- This is followed by the same determination in a fresh sample of the same plant or animal species.
- Then the ^{14}C content of the old and fresh samples of the organisms are compared. Knowing the half life of ^{14}C , the age of the old material can be estimated.



Problem 3

- The measurement of the radioactivity of a piece of wood showed a disintegration rate of 6.12 counts per minute by a gram of carbon. A fresh sample of wood showed the rate of disintegration of 15.3 per minute per gram of carbon. Assuming the ratio of ^{14}C to ^{12}C in the atmosphere remains constant, calculate the age of the old plant material. The half life of ^{14}C is 5730 years.

- **Solution**

- Being a first order reaction, the disintegration will be proportional to the concentration. We can use the integrated equation for first order reactions.

$$\ln \frac{[A]_0}{[A]_t} = kt$$

$$\ln \frac{15.3}{6.12} = kt$$



- Given the half life of ^{14}C as 5730 years, we can find k from the relation,

$$\begin{aligned}k &= \frac{0.693}{5730} \\&= 1.2 \times 10^{-4} \text{ yr}^{-1} \\&= \ln \frac{15.3}{6.12} = t \times 1.2 \times 10^{-4} \\&= 7576 \text{ years}\end{aligned}$$

- From the answer it can be seen that the ^{14}C in the wood has decayed through more than one half life as the disintegration rate has decreased from 15.3 to 6.12.



Problem 4

- A 1.0 gram sample of a bone from an excavation in an abandoned cemetery gave 7900 ^{14}C disintegrations in twenty-four hours. Within the same period, 1 gram of carbon from a fresh sample of bone underwent 18,400 disintegration. Calculate the age of the old bone sample.

- **Solution**

- As in the last problem, k can be worked out to be $1.2 \times 10^{-4} \text{ yr}^{-1}$. If we represent the initial and final disintegrations to be represented by N_0 and N respectively, then
 - $\ln \frac{N_0}{N} = kt$
 - $\ln \frac{18,400}{7,900} = t \times 1.2 \times 10^{-4}$
 - $t = 7075 \text{ years}$

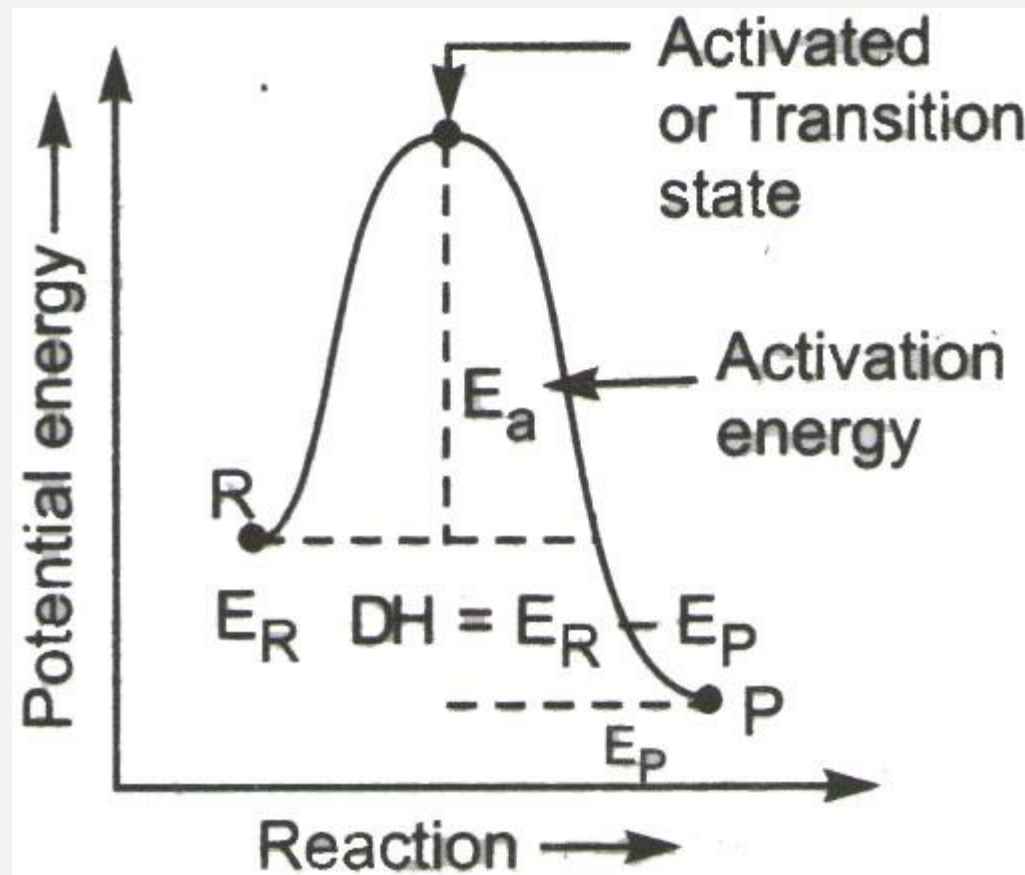


Effect of temperature on the rate of reaction

- For most chemical reactions, increasing temperature of reaction by 10°C at least doubles the rate of reaction.
- As temperature is increased, the kinetic energy of the reacting molecules increases, and consequently, collision frequency.
- However, collision per se cannot account for the doubling of the reaction rate.
- Not all the collisions can result in reaction.
- The observations that not all collision can result in reaction, and that reaction rate increases with temperature increase, can be accounted for by the fact that a certain minimum energy is required for reaction to take place.



- That minimum energy required for reaction is called the activation energy (E_a)



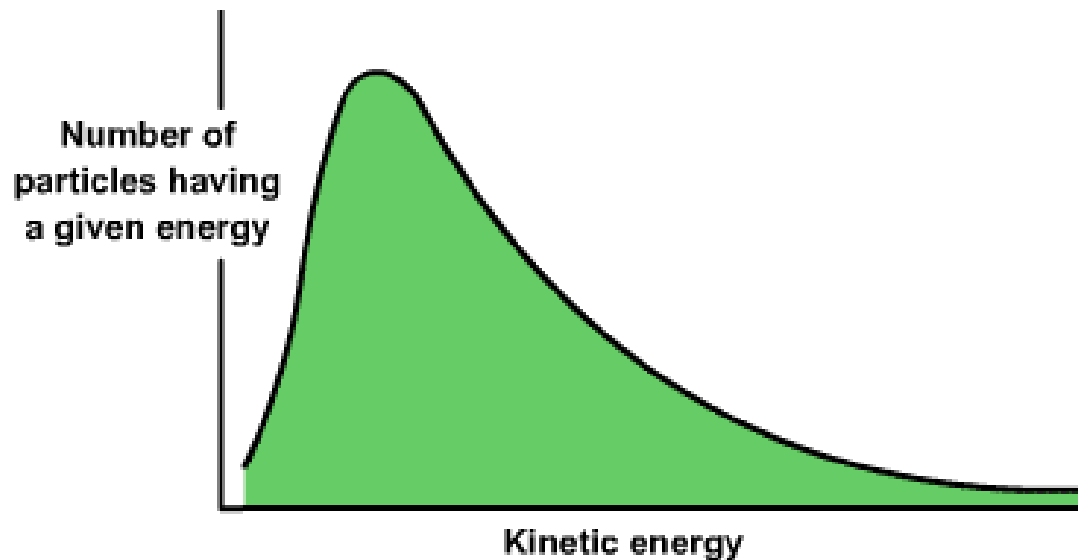
- For the activated complex to be formed, there is an energy barrier that has to be overcome-this is the activation energy.
- The existence of the energy barrier is attributed to the repulsion between reacting molecules.
- Additionally, bond breaking requires input of energy.
- The repulsion is overcome by the kinetic energy of the approaching molecules.



- When the reactants attain the activation energy, they are able to overcome the repulsive forces and make effective collisions.
- That energy enables bonds to be broken, and also ensures rearrangement of atoms, ions and electrons as reaction proceeds.
- The magnitude of E_a determines how fast a reaction will be.
- If E_a is high, only a small proportion of reacting molecules have sufficient energy to overcome the energy barrier, so reaction rate is low.
 - On the contrary, if the E_a is low, then a larger proportion of the reacting molecules can attain this energy to make effective collisions.

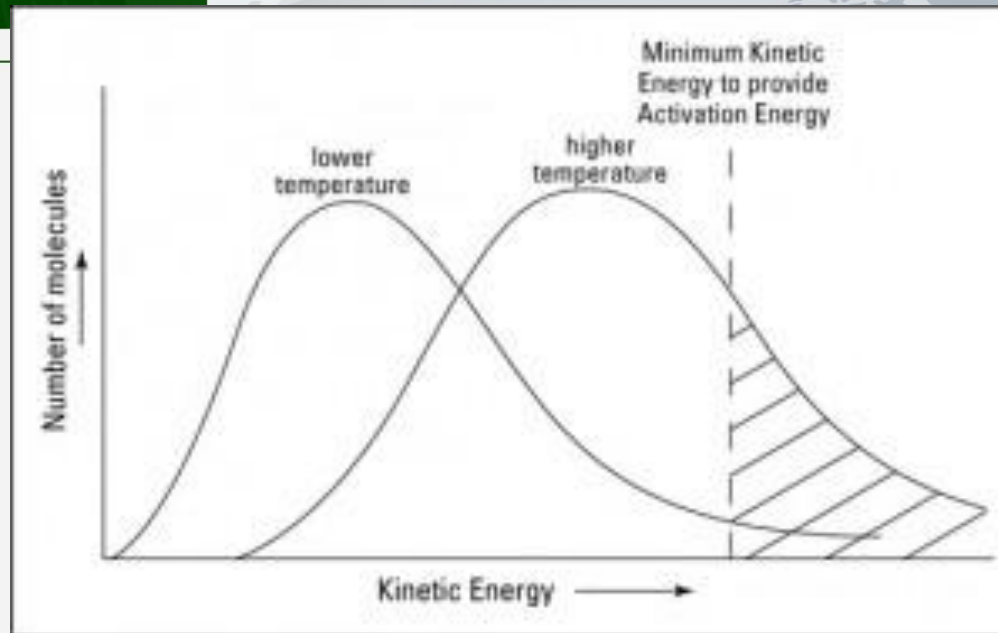


- To explain how a 10° rise in temperature affects the rate of reaction, Maxwell and Boltzmann used probability theory to show how energy is distributed among gaseous molecules.
- This form of distribution is shown on a graph in which number of particles with certain energy is plotted against kinetic energy.



- The area under the curve is proportional to the total number of particles involved in the reaction.
- Only a small proportion of the molecules have energies equal to or greater than the activation energy.
- This number of molecules is proportional to the shaded area.
- The majority of molecules have energies less than E_a .
- The second graph below shows the distribution of energies at two different temperatures, T_1 and T_2 where T_2 is greater than T_1 .





- As the temperature is raised, the curve is flattened and shifted to the right.
- For the sample of gaseous reactants, the areas under the curves T1 and T2 are equal.
- But the shaded area under the curve T2 is larger than the corresponding area under T1



- For a 10-degree difference between T1 and T2, this will be reflected in about two times increase in the shaded area of the curve for the higher temperature.
- Maxwell and Boltzmann derived equations for the distribution of energies amongst molecules.
- They proposed that the fraction of molecules with energy greater than E_a is given by $e^{-E_a/RT}$ where R is the molar gas constant, T is the absolute temperature and e is the base of natural log.
- To take care of molecular collisions, the following equation was proposed;
- $k = Ze^{-E_a/RT}$
 - where k is the rate constant, Z is the collision number or the total number of collisions per second.

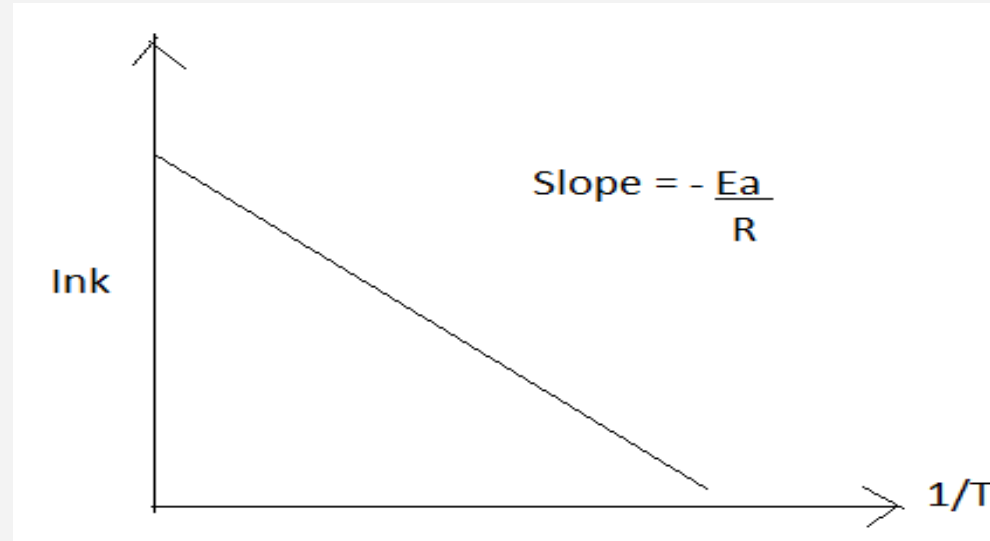


- Not all molecules attaining E_a will necessarily react with each other as the factor of orientation amongst the molecules is also important.
- In order to cater for the orientation of the molecules, the equation was modified to $k = PZe^{E_a/RT}$
 - Where P stands for the probability or steric factor.
- There was further modification of the last equation by Arrhenius. His equation was $k = Ae^{-E_a/RT}$.
- A is the Arrhenius constant, a pre-exponential factor which stands for the collision frequency and the orientation of the colliding molecules.



- Transforming the Arrhenius equation to the natural log (ln) form gives $\ln k = \ln A - \frac{E_a}{RT}$

Or $\log_{10} k = \log_{10} A - \frac{E_a}{2.30RT}$



- The last two equations are in the form of an equation for a straight line; $y = mx + c$
- When $\ln k$ or $\log_{10} k$ is plotted against $1/T$, a straight line is obtained. The slope of the line is $-E_a/R$ or $-E_a/2.30R$.
- The intercept on the y-axis is $\ln A$ or $\log_{10} A$.



- The E_a of a reaction can be determined by the measurements of different rates and hence different rate constants at different temperatures.
- A graph of $\ln k$ is then plotted against $1/T$.
- An alternative method of determining E_a is to measure rates and for that matter rate constants at two different temperatures, T_1 and T_2 .
- Then use the following equation to determine the only unknown, E_a
- $\ln k_1/k_2 = E_a/R(1/T_1 - 1/T_2)$

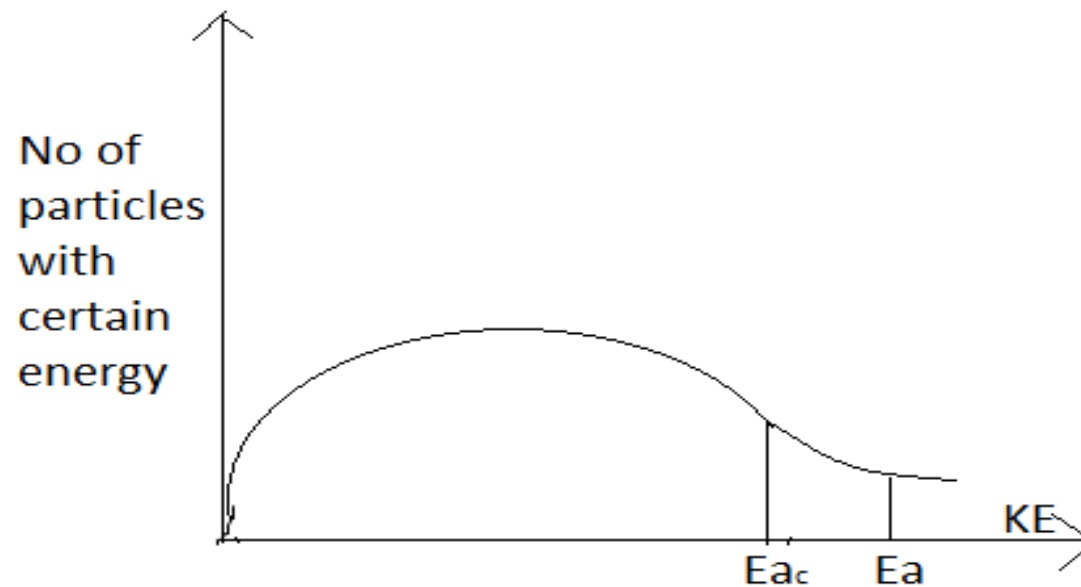


Catalysis

- A catalyst is a substance that speeds up chemical reactions without undergoing any permanent chemical change.
- A catalyst provides an alternative pathway of lowered activation energy so that more reacting molecules are able to attain this energy to undergo reaction.
- In the presence of a catalyst, a greater traction of reacting molecules have sufficient energy to overcome the lowered energy barrier.
- A catalyst does not supply energy to reactants as the energy content of the reactants and products remain the same.



- So a catalyst has no effect on the enthalpy change or free energy change of reaction.
- Neither does a catalyst initiate reactions .
- It quickens thermodynamically feasible reactions.



- E_a in the absence of a catalyst is higher than E_{ac} in the presence of a catalyst.



Homogenous and heterogenous catalysis

- If a catalyst is present in the same phase as the reactant then it is called a homogenous catalyst.
- Few drops of liquid bromine can be added to liquid hydrogen peroxide to increase the rate of decomposition of the peroxide.
- This is an example of homogenous catalysis.
- If the phase in which the catalyst occurs is different from the phase of the reactant, the catalyst is a heterogenous one.
- The rate of decomposition of hydrogen peroxide can be quickened by adding MnO_2 in solid form.
- In this instance the Manganese IV oxide is as a heterogenous catalyst.



- The most common heterogeneous catalysts are solids that are used in gas-phase or liquid-phase reactions.
- For example in the Haber process, iron catalyst is used for the reaction between H_2 and N_2 gases to form ammonia.
- Then in the contact process for the production of H_2SO_4 , the reaction between gaseous SO_2 and O_2 is catalysed by finely divided vanadium V oxide, V_2O_5 .



Some characteristics of catalysts

- They increase rate of reactions
- They are used in small quantities
- They may take part in the reaction, but are regenerated at the end of the reaction
- After catalysis, their chemical nature does not change, but their physical state could change
- Catalysts are specific: however, they are less specific than enzymes
- Catalysts can be poisoned in which case the effectiveness of the catalyst is reduced or inhibited.



Theories explaining reaction rates

Collision theory

- According to this theory, reactions take place between reactants if the particles involved collide, and if the particles have the requisite energy.
- Not all collisions are effective as the reacting particles have to meet at a particular angle.
- From the collision theory, the rate of any step in a reaction is directly proportional to
 - i) the number of collisions per second between the particles.
 - ii) The fraction of the collision which is effective.



- Take the reaction between X and Y molecules to form Z molecules.
- $X + Y \rightarrow Z$
- In the reaction vessel, if the concentration of X and Y is increased, then the collision frequency between X and Y will also increase, so there will be increased reaction rate.
- Not all collisions are effective to result in reaction. Some of the collisions are so gentle that there is no change in the identity of the molecules after collision.
- Such collisions are said to be perfectly elastic. In a gentle collision, the repulsion between the electron clouds may cause the molecules to bounce apart.



- However, if X and Y have enough kinetic energy before collision, they can use this kinetic energy to overcome the repulsive forces.
- It is where the reacting molecules have energies at least equal to the activation energy that the repulsive forces can be overcome for effective collisions to be made.
- The magnitude of E_a depends on the nature of reactants.
- Collision theory can account for all the factors that influence reaction rate.
 - Nature of reactants
 - Temperature
 - Concentration of reactants
 - Catalyst.



The transition state theory

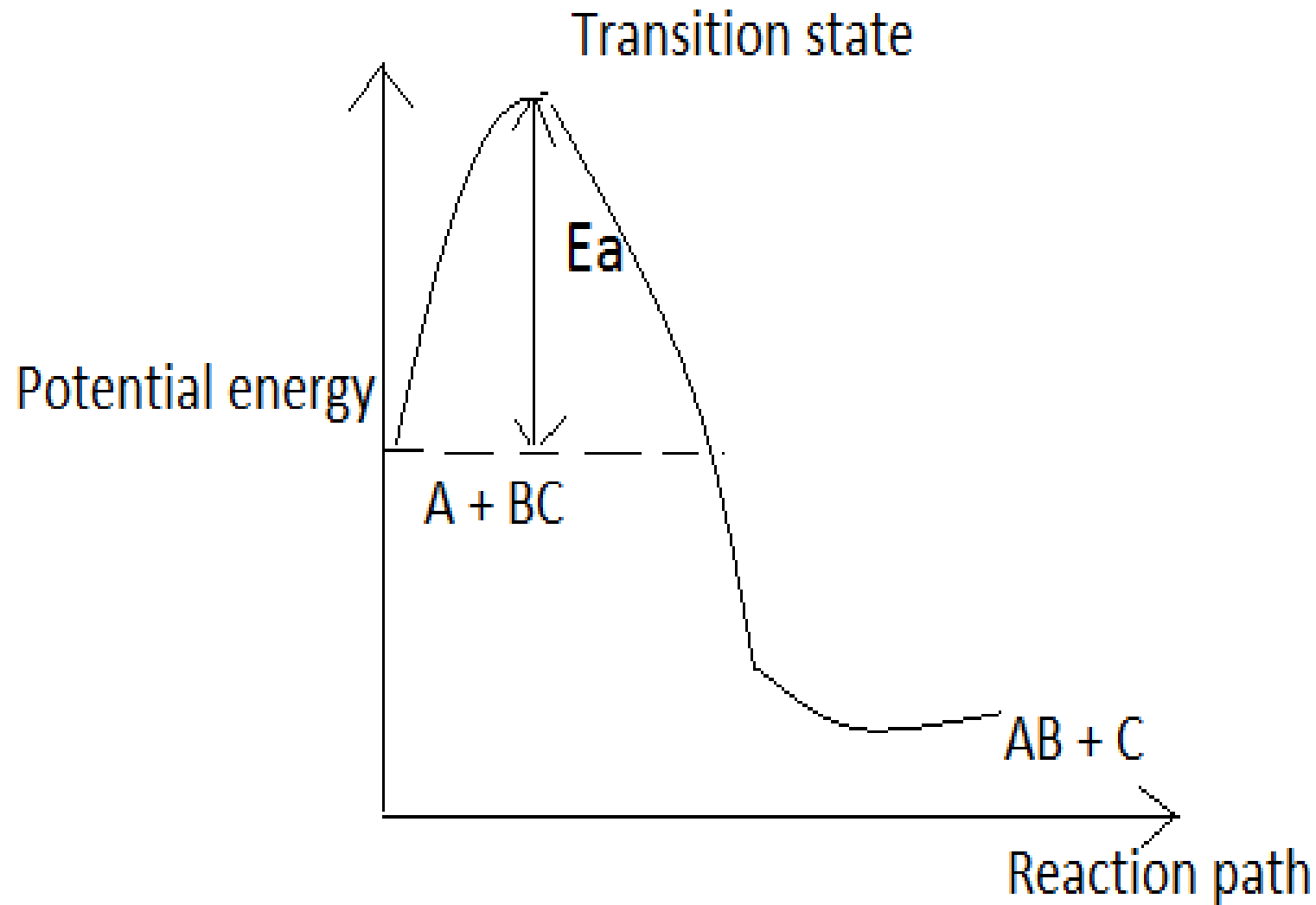
- Take the reaction; $A + BC \rightarrow AB + C$, where BC and AB are diatomic molecules while A and C are atoms.
- In the context of the transition state theory, the above reaction can be envisaged to involve the following steps.
- **Step I:** A and BC approach each other.
- **Step II:** Rearrangement of A and BC to form a transition state in which old bonds are partially broken, and new bonds partially formed
- **Step III:** Either new products, AB and C are formed from the transition state or reactants, A and BC are reformed.



$A + BC$
Reactants

$A \cdots B \cdots C$
transition state

$AB + C$
products



■ Step I

- As A and BC approach each other, repulsive forces arise mainly from the electron cloud of the molecules.
- A minimum amount of energy, E_a , is required to overcome the repulsive forces from A and BC to enable them approach closely enough for reaction to occur.
- This energy is provided by the kinetic energy of the reacting molecules.
- As A and BC get closer, they slow down.
- There is decrease in the kinetic energy while the potential energy of the system increases (note that energy cannot be lost).



- If the kinetic energy of reactants is smaller than E_a , the reactants cannot approach close enough to attain the potential energy arrangement in step II, so the collision of the less energetic molecules would be elastic.
- But if the kinetic energy is equal to, or greater than E_a , the transition state is formed.
- **Step II**
 - Part or all the kinetic energy the reactants had before collision is converted into the potential energy of the transition state.
 - This state is unstable and can decompose to give either products or reactants.
- **Step III**
 - If products are formed from the activated complex, reaction has occurred, and as products separate, potential energy is converted to kinetic energy.



Comparison of the collision and transition state theories.

- In the transition state theory, the emphasis is on the transition state, and it is the potential energy of the system which is considered but not the kinetic energy.
- That is why it is the potential energy which is shown on the energy profile diagram.
- The main significance of the transition state theory is that it provides a means of picturing the activated complex in terms of modern concepts on bonding and molecular structures.
- However, this theory does not give any quantitative measure of the activation energy.



- On the other hand, the collision theory is rather silent on how old bonds are broken and new ones formed.
- It focuses on the events culminating in collision and reaction.
- It also gives a quantitative estimate of the factors involved in collision, such as collision frequency, orientation of colliding molecules, as reflected in the Arrhenius equation.



Energy profile diagram for a two-step reaction

- A two-step reaction is an example of a complex reaction which involves more than one step and it is also characterized by the formation of an intermediate
- The overall energy diagram represents the profiles of two reactions, with the intermediate serving as a link between the two steps.
- Where such an intermediate is formed, there is a minimum in the energy profile diagram.
- The deeper the drop, the more stable the intermediate would be.

There are two transition states, TS_1 and TS_2 but only one intermediate.



potential energy

